MA-310 (Calculus III)

Worcester State University

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Contents

Chapter 1

Vectors and the Geometry of Space

Vector calculus is the branch of mathematics that deals with vector fields and their derivatives. It is essential for understanding the behavior of physical systems in three-dimensional space.

In this chapter, we will explore the fundamental concepts of vector calculus, including gradient, divergence, and curl.

1.1 Vectors in Two Dimensions

A vector is a quantity that has both magnitude and direction. In two dimensions, vectors can be represented as arrows in a plane, where the length of the arrow corresponds to the magnitude of the vector, and the direction of the arrow indicates the direction of the vector.

Definition 1.1 A velocity or force has both magnitude and direction and is often represented by a **directed line segment** or a **vector**. We denote the vector with **initial point** P and **terminal point** Q as $\vec{PQ}$. The direction of the vector indicates arrowhead at Q. The **magnitude** of $\vec{PQ}$ is the length of the segment PQ and is denoted by $\|\vec{PQ}\|$. Vectors that have the same magnitude and direction are called **equivalent**. ◆

A vector may be translated from one location to another, provided neither the magnitude nor the direction is changed. If $\vec{PS} = \vec{PQ} + \vec{PR}$ and $\vec{PQ}$ and $\vec{PR}$ are two forces acting at P, then $\vec{PS}$ is the **resultant force** (sum).

If c is a scalar and $\vec{v}$ is a vector, then $c\vec{v}$ (a **scalar multiple**) is defined as a vector whose magnitude is $|c|$ times the magnitude $\|\vec{v}\|$ of $\vec{v}$ with the same direction ($c > 0$) or the opposite direction ($c < 0$) of $\vec{v}$.

A vector with the initial point at the origin is called the **position vector**. We use the symbol $\langle a_1, a_2 \rangle$ for an ordered pair that represents the vector and denote it by a lowercase letter $\vec{v} = \langle a_1, a_2 \rangle$.

Two vectors $\langle a_1, a_2 \rangle$ and $\langle b_1, b_2 \rangle$ are **equal** $\iff$ $a_1 = b_1$ and $a_2 = b_2$.

Example 1.2 Sketch the position vectors for:

1 $\vec{a} = \langle -3, 2 \rangle$

2 $\vec{b} = \langle 0, -2 \rangle$

3 $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$

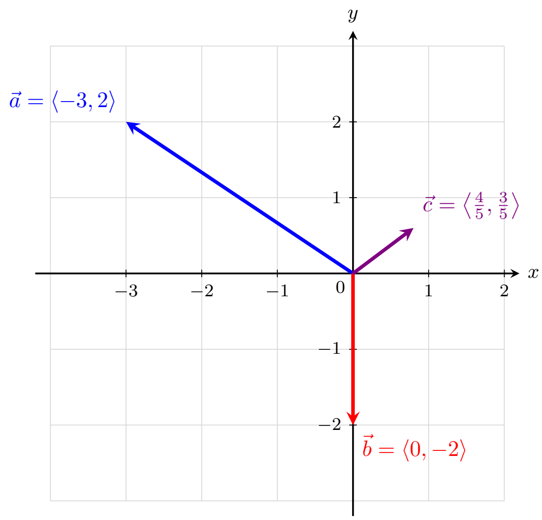


Figure 1.3 The position vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ drawn in the Cartesian plane.

Solution. To sketch a position vector, always place its initial point at the origin $(0, 0)$. The terminal point will match the components of the vector:

1 For $\vec{a} = \langle -3, 2 \rangle$, draw an arrow starting at $(0, 0)$ and ending at the point $(-3, 2)$.

2 For $\vec{b} = \langle 0, -2 \rangle$, draw an arrow starting at $(0, 0)$ and pointing straight down to end at the point $(0, -2)$.

3 For $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$, draw an arrow starting at $(0, 0)$ and ending at the point $(0.8, 0.6)$ in the first quadrant.

▲

Definition 1.4 The **magnitude** (or length) $\|\vec{a}\|$ of the vector $\vec{a} = \langle a_1, a_2 \rangle$ is

$$\|\vec{a}\| = \|\langle a_1, a_2 \rangle\| = \sqrt{a_1^2 + a_2^2}.$$

◆

Example 1.5 Find the magnitude of each vector:

1 $\vec{a} = \langle -3, 2 \rangle$

2 $\vec{b} = \langle 0, -2 \rangle$

3 $\vec{c} = \left\langle \dfrac{4}{5}, \dfrac{3}{5} \right\rangle$

Solution. We apply the magnitude formula $\|\vec{v}\| = \sqrt{v_1^2 + v_2^2}$ to each vector:

1 For $\vec{a} = \langle -3, 2 \rangle$:

$$\|\vec{a}\| = \sqrt{(-3)^2 + 2^2} = \sqrt{9 + 4} = \sqrt{13}$$

2 For $\vec{b} = \langle 0, -2 \rangle$:

$$\|\vec{b}\| = \sqrt{0^2 + (-2)^2} = \sqrt{0 + 4} = \sqrt{4} = 2$$

3 For $\vec{c} = \left\langle \frac{4}{5}, \frac{3}{5} \right\rangle$:

$$\|\vec{c}\| = \sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{3}{5}\right)^2} = \sqrt{\frac{16}{25} + \frac{9}{25}} = \sqrt{\frac{25}{25}} = 1$$

Because its magnitude is 1, $\vec{c}$ is a unit vector.

▲

Fact 1.6 Vector Operations. ***Addition of vectors***

$$\langle a_1, a_2 \rangle + \langle b_1, b_2 \rangle = \langle a_1 + b_1, a_2 + b_2 \rangle$$

Scalar multiple of a vector

$$c\langle a_1, a_2 \rangle = \langle ca_1, ca_2 \rangle$$

Example 1.7 Let $\vec{a} = \langle -1, 3 \rangle$ and $\vec{b} = \langle 4, 3 \rangle$.

1 Find $\vec{a} + \vec{b}$ and $-3\vec{a} + 2\vec{b}$.

2 Sketch vectors $\vec{a} + \vec{b}$ and $-\frac{3}{2}\vec{b}$.

Solution.

1 To find $\vec{a} + \vec{b}$, add the corresponding components:

$$\vec{a} + \vec{b} = \langle -1 + 4, 3 + 3 \rangle = \langle 3, 6 \rangle$$

To find $-3\vec{a} + 2\vec{b}$, apply scalar multiplication first, then add:

$$-3\vec{a} = -3\langle -1, 3 \rangle = \langle 3, -9 \rangle$$

$$2\vec{b} = 2\langle 4, 3 \rangle = \langle 8, 6 \rangle$$

$$-3\vec{a} + 2\vec{b} = \langle 3 + 8, -9 + 6 \rangle = \langle 11, -3 \rangle$$

2 First, calculate the scalar multiple: $-\frac{3}{2}\vec{b} = -\frac{3}{2}\langle 4, 3 \rangle = \left\langle -6, -\frac{9}{2} \right\rangle = \langle -6, -4.5 \rangle$.

Below is the sketch of the two position vectors on the coordinate plane:

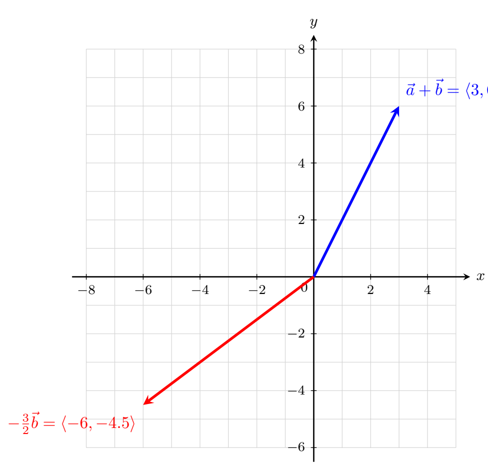


Figure 1.8 The vectors $\vec{a} + \vec{b}$ and $-\frac{3}{2}\vec{b}$.

▲

Definition 1.9 The **zero vector 0** is defined as $\vec{0} = \langle 0, 0 \rangle$, and the **negative -a** of a vector $\vec{a} = \langle a_1, a_2 \rangle$ is defined as:

$$-\vec{a} = -\langle a_1, a_2 \rangle = \langle -a_1, -a_2 \rangle$$

◆

Theorem 1.10 Properties of Vectors. *Let $\vec{a}$, $\vec{b}$, and $\vec{c}$ be vectors, and let c and d be scalars. Then:*

1 $\vec{a} + \vec{b} = \vec{b} + \vec{a}$, *and* $c(\vec{a} + \vec{b}) = c\vec{a} + c\vec{b}$

2 $\vec{a} + (\vec{b} + \vec{c}) = (\vec{a} + \vec{b}) + \vec{c}$

3 $\vec{a} + \vec{0} = \vec{a}$, *and* $\vec{a} + (-\vec{a}) = \vec{0}$

4 $(c + d)\vec{a} = c\vec{a} + d\vec{a}$, *and* $(cd)\vec{a} = c(d\vec{a}) = d(c\vec{a})$

5 $1\vec{a} = \vec{a}$, *and* $0\vec{a} = \vec{0} = c\vec{0}$

Definition 1.11 Let $\vec{a} = \langle a_1, a_2 \rangle$ and $\vec{b} = \langle b_1, b_2 \rangle$. The **difference a-b** is:

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}) = \langle a_1 - b_1, a_2 - b_2 \rangle$$

◆

Example 1.12 If $\vec{a} = \langle 5, -3 \rangle$ and $\vec{b} = \langle -2, 3 \rangle$, find:

1 $\vec{a} - \vec{b}$

2 $3\vec{a} - 2\vec{b}$

Solution.

1 To find $\vec{a} - \vec{b}$, subtract the components of $\vec{b}$ from the components of $\vec{a}$:

$$\vec{a} - \vec{b} = \langle 5 - (-2), -3 - 3 \rangle = \langle 5 + 2, -6 \rangle = \langle 7, -6 \rangle$$

2 To find $3\vec{a} - 2\vec{b}$, perform the scalar multiplications first, then subtract:

$$3\vec{a} = 3\langle 5, -3 \rangle = \langle 15, -9 \rangle$$

$$2\vec{b} = 2\langle -2, 3 \rangle = \langle -4, 6 \rangle$$

Now subtract the resulting components:

$$3\vec{a} - 2\vec{b} = \langle 15 - (-4), -9 - 6 \rangle = \langle 15 + 4, -15 \rangle = \langle 19, -15 \rangle$$

▲

Theorem 1.13 Vector from Two Points. *If $P_1(x_1, y_1)$ and $P_2(x_2, y_2)$ are any two points, the vector $\vec{a}$ in V_2 that corresponds to the directed line segment $\vec{P_1P_2}$ is:*

$$\vec{a} = \langle x_2 - x_1, y_2 - y_1 \rangle$$

Example 1.14 Given points $P(3, 2)$ and $Q(-1, 5)$:

1 Sketch $\vec{PQ}$ and $\vec{QP}$.

2 Find vectors $\vec{a}$ and $\vec{b}$ in V_2 that correspond to $\vec{PQ}$ and $\vec{QP}$.

3 Sketch the position vectors for $\vec{a}$ and $\vec{b}$.

Solution.

1 To sketch the directed line segments, draw an arrow starting at point $P(3,2)$ and ending at point $Q(-1,5)$ for $\vec{PQ}$. For $\vec{QP}$, reverse the direction so it starts at $Q(-1,5)$ and ends at $P(3,2)$. (See the left graph below).

2 To find the vector components, subtract the initial coordinates from the terminal coordinates ("terminal minus initial"):

For $\vec{a}$ corresponding to $\vec{PQ}$:

$$\vec{a} = \langle -1-3, 5-2\rangle = \langle -4, 3\rangle$$

For $\vec{b}$ corresponding to $\vec{QP}$:

$$\vec{b} = \langle 3-(-1), 2-5\rangle = \langle 4, -3\rangle$$

3 To sketch the position vectors $\vec{a}$ and $\vec{b}$, shift them so their initial points sit exactly at the origin $(0,0)$. The terminal points will be $(-4,3)$ and $(4,-3)$, respectively. Notice that they are perfectly parallel to and have the same lengths as the original segments (See the right graph below).

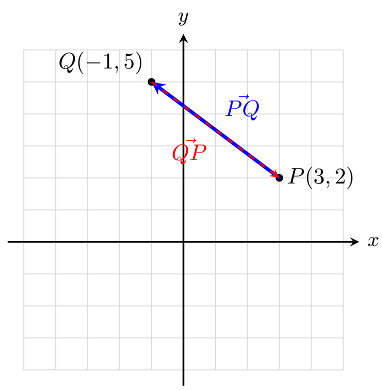


Figure 1.15 The directed line segments $\vec{PQ}$ and $\vec{QP}$ in space.

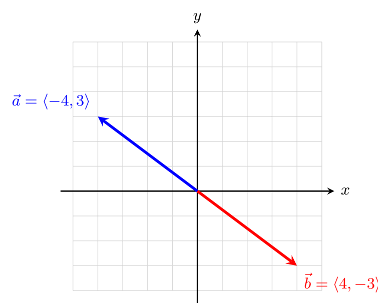


Figure 1.16 The corresponding position vectors $\vec{a}$ and $\vec{b}$ at the origin.

▲

Definition 1.17 Nonzero vectors $\vec{a}$ and $\vec{b}$ in V_2 have:

1 the **same direction** if $\vec{b} = c\vec{a}$ for some scalar $c > 0$

2 the **opposite direction** if $\vec{b} = c\vec{a}$ for some scalar $c < 0$

◆

Theorem 1.18 Magnitude of a Scalar Multiple. *If $\vec{a}$ is a vector and c is a scalar, then $\|c\vec{a}\| = |c|\|\vec{a}\|$.*

Definition 1.19 A vector of length 1 is called a **unit vector**. The **standard unit vectors** are:

$$\vec{i} = \langle 1, 0\rangle, \quad \vec{j} = \langle 0, 1\rangle$$

If $\vec{a} = \langle a_1, a_2\rangle$, then it can be written as a linear combination of the standard unit vectors:

$$\vec{a} = a_1\vec{i} + a_2\vec{j}$$

◆

Example 1.20 If $\vec{a} = -5\vec{i} + 2\vec{j}$ and $\vec{b} = \vec{i} - 3\vec{j}$, express $2\vec{a} - 3\vec{b}$ as a linear combination of $\vec{i}$ and $\vec{j}$.

Solution. Substitute the linear combinations for $\vec{a}$ and $\vec{b}$ into the expression, distribute the scalars, and combine like terms:

$$\begin{aligned} 2\vec{a} - 3\vec{b} &= 2(-5\vec{i} + 2\vec{j}) - 3(\vec{i} - 3\vec{j}) \\ &= (-10\vec{i} + 4\vec{j}) - (3\vec{i} - 9\vec{j}) \\ &= -10\vec{i} - 3\vec{i} + 4\vec{j} + 9\vec{j} \\ &= -13\vec{i} + 13\vec{j} \end{aligned}$$

▲

Theorem 1.21 Unit Vector in the Same Direction. *A **unit vector** is a vector of magnitude 1. The vectors $\vec{i}$ and $\vec{j}$ are unit vectors. If $\vec{a} \neq \vec{0}$, then the unit vector $\vec{u}$ that has the same direction as $\vec{a}$ is:*

$$\vec{u} = \frac{1}{\|\vec{a}\|}\vec{a}$$

Example 1.22 Find the unit vector $\vec{u}$ that has the same direction as $\vec{a} = -8\vec{i} + 15\vec{j}$.

Solution. First, find the magnitude of vector $\vec{a}$:

$$\|\vec{a}\| = \sqrt{(-8)^2 + 15^2} = \sqrt{64 + 225} = \sqrt{289} = 17$$

Next, multiply the reciprocal of the magnitude by the original vector $\vec{a}$:

$$\vec{u} = \frac{1}{17}(-8\vec{i} + 15\vec{j}) = -\frac{8}{17}\vec{i} + \frac{15}{17}\vec{j}$$

▲

Example 1.23 A quarterback releases the football with a velocity of 50 ft/sec at an angle of 35° with the horizontal. Approximate the horizontal and vertical components of the vector.

Solution. A vector $\vec{v}$ given a magnitude $\|\vec{v}\|$ and a direction angle θ can be written in component form as $\vec{v} = \langle \|\vec{v}\| \cos(\theta), \|\vec{v}\| \sin(\theta) \rangle$.

Given $\|\vec{v}\| = 50$ and $\theta = 35°$:

- Horizontal component: $50\cos(35°) \approx 40.96$ ft/sec
- Vertical component: $50\sin(35°) \approx 28.68$ ft/sec

In component form, the velocity vector is approximately $\langle 40.96, 28.68 \rangle$. ▲

Example 1.24 A jet airplane approaches a runway at an angle of 7.5° with the horizontal, traveling at a velocity of 160 mi/hr. Approximate the horizontal and vertical components of the vector.

Solution. Given a magnitude of $\|\vec{v}\| = 160$ and an approach angle of $\theta = 7.5°$:

- Horizontal component: $160\cos(7.5°) \approx 158.63$ mi/hr
- Vertical component: $160\sin(7.5°) \approx 20.88$ mi/hr

In component form, the velocity vector is approximately $\langle 158.63, 20.88 \rangle$. Note that because the plane is descending, the vertical velocity component can be interpreted as acting in the downward direction (-20.88). ▲

1.2 Vectors in Three Dimensions

In order to represent points in space, we first choose a fixed point O (the origin) and three directed lines through O which are perpendicular to each other, called the **coordinate axes**. They are labeled as the x**-axis**, y**-axis**, and z**-axis**. Think of the x- and y-axes as being horizontal while the z-axis is vertical. The direction of the z-axis is determined by the **right-hand rule**.

The three coordinate axes determine the three **coordinate planes**. The xy-plane contains the x- and y-axes, the yz-plane contains the y- and z-axes, and the xz-plane contains the x- and z-axes. These three coordinate planes divide space into eight parts, called **octants**. The **first octant** is determined by the positive axes.

We represent a point P by the ordered triple (a, b, c) of real numbers and we call a, b, and c the **coordinates** of P; here, a, b, and c are the x-, y-, and z-coordinates, respectively.

The point $P(a, b, c)$ can be visualized as a corner of a rectangular box. If we drop a perpendicular from P to the xy-plane, we get a point Q with coordinates $(a, b, 0)$ called the **projection** of P on the xy-plane.

The Cartesian product $\mathbb{R} \times \mathbb{R} \times \mathbb{R} = \{(x, y, z) \mid x, y, z \in \mathbb{R}\}$ is the set of all ordered triples of real numbers and is denoted by $\mathbb{R}^3$.

Example 1.25 What surfaces in $\mathbb{R}^3$ are represented by the following equations?

1 $z = 3$

2 $y = 5$

Solution.

1 The equation $z = 3$ represents the set of all points $\{(x, y, z) \mid z = 3\}$ in $\mathbb{R}^3$. Since x and y can be any real numbers, this forms a horizontal plane parallel to the xy-plane, located exactly 3 units above it.

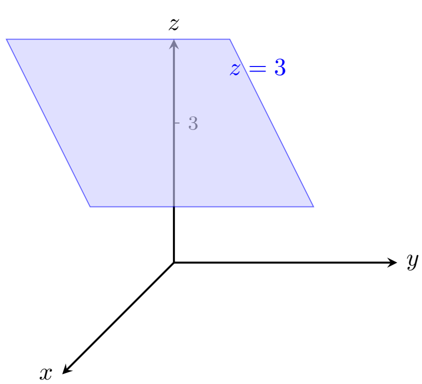


Figure 1.26 The plane $z = 3$ in $\mathbb{R}^3$.

2 The equation $y = 5$ represents the set of all points $\{(x, y, z) \mid y = 5\}$ in $\mathbb{R}^3$. Since x and z can be any real numbers, this forms a vertical plane parallel to the xz-plane, located exactly 5 units to the right of it.

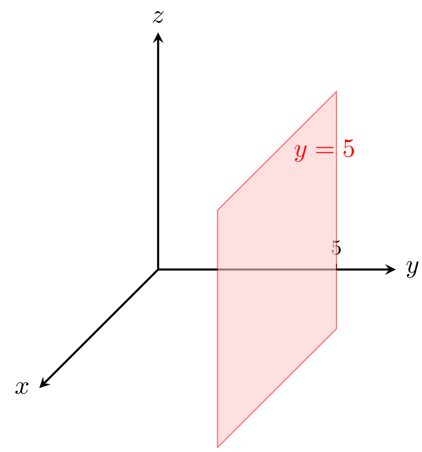


Figure 1.27 The plane $y = 5$ in $\mathbb{R}^3$.

▲

Fact 1.28 Distance Formula in Three Dimensions. *The distance* $d(P_1, P_2) = |P_1P_2|$ *between the points* $P_1(x_1, y_1, z_1)$ *and* $P_2(x_2, y_2, z_2)$ *is:*

$$d(P_1, P_2) = |P_1P_2| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$$

Example 1.29 Find the distance from the point $P(-1, -3, 1)$ to the point $Q(3, 4, -2)$.

Solution. Substitute the coordinates of P and Q into the three-dimensional distance formula:

$$\begin{aligned} |PQ| &= \sqrt{(3 - (-1))^2 + (4 - (-3))^2 + (-2 - 1)^2} \\ &= \sqrt{(3 + 1)^2 + (4 + 3)^2 + (-3)^2} \\ &= \sqrt{4^2 + 7^2 + (-3)^2} \\ &= \sqrt{16 + 49 + 9} \\ &= \sqrt{74} \end{aligned}$$

The exact distance between the points is $\sqrt{74}$. ▲

Fact 1.30 Equation of a Sphere. *An equation of a sphere with center* $C(h, k, l)$ *and radius* r *is:*

$$(x - h)^2 + (y - k)^2 + (z - l)^2 = r^2$$

If the center is at the origin $O(0, 0, 0)$, *then the equation of the sphere simplifies to:*

$$x^2 + y^2 + z^2 = r^2$$

Example 1.31 Show that $x^2 + y^2 + z^2 + 4x - 6y + 2z + 6 = 0$ is the equation of a sphere, and find its center and radius.

Solution. To transform the given equation into standard form, group the variables and complete the square for each variable:

$$\begin{aligned} (x^2 + 4x) + (y^2 - 6y) + (z^2 + 2z) &= -6 \\ \left(x^2 + 4x + \left(\frac{4}{2}\right)^2\right) + \left(y^2 - 6y + \left(\frac{-6}{2}\right)^2\right) + \left(z^2 + 2z + \left(\frac{2}{2}\right)^2\right) &= -6 + 4 + 9 + 1 \\ (x^2 + 4x + 4) + (y^2 - 6y + 9) + (z^2 + 2z + 1) &= 8 \end{aligned}$$

$$(x+2)^2 + (y-3)^2 + (z+1)^2 = 8$$

Comparing this result to the standard equation of a sphere $(x-h)^2 + (y-k)^2 + (z-l)^2 = r^2$, we find:

- The center of the sphere is $C(h,k,l) = C(-2,3,-1)$.
- The radius is $r = \sqrt{8} = 2\sqrt{2}$.

▲

Example 1.32 What region in $\mathbb{R}^3$ is represented by the following inequalities?

$$1 \leq x^2 + y^2 + z^2 \leq 4$$

Solution. The expression $x^2 + y^2 + z^2 = r^2$ represents the squared distance from the origin to a point in three-dimensional space. We can rewrite the inequality by taking the square root of all parts:

$$1 \leq \sqrt{x^2 + y^2 + z^2} \leq 2$$

This represents the set of all points whose distance r from the origin satisfies $1 \leq r \leq 2$. Therefore, the region is a solid **spherical shell** (or a thick hollow sphere) centered at the origin, including the inner sphere of radius 1 and the outer sphere of radius 2.

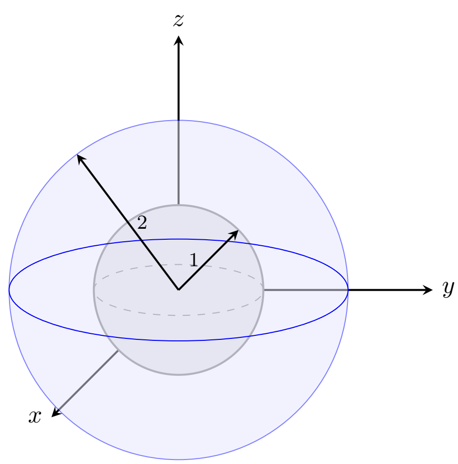


Figure 1.33 The spherical shell bounded by $r = 1$ and $r = 2$.

▲

Example 1.34 What region in $\mathbb{R}^3$ is represented by the following inequalities?

$$x^2 + y^2 \leq 25, \quad |z| \leq 3$$

Solution. We analyze the two conditions independently to find their intersection:

- In the xy-plane, $x^2 + y^2 \leq 25$ describes a solid disk of radius 5 centered at the origin. Since z is unconstrained by this expression, it forms an infinitely long solid cylinder extending along the z-axis.
- The absolute value inequality $|z| \leq 3$ is equivalent to $-3 \leq z \leq 3$. This bounds the space vertically between two horizontal planes.

Combining these conditions restricts the infinite cylinder to a specific height. Therefore, the region is a **solid vertical cylinder** of radius 5 and height 6, centered along the z-axis and extending from $z = -3$ to $z = 3$.

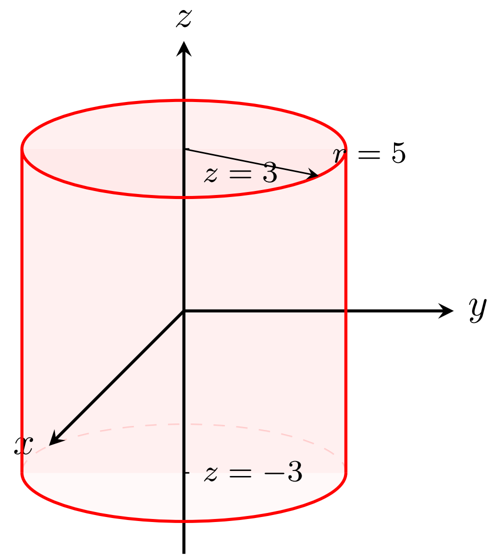


Figure 1.35 The solid vertical cylinder bounded by $x^2 + y^2 \leq 25$ and $-3 \leq z \leq 3$.

▲

Fact 1.36 Vector Operations in Three Dimensions. *Let $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ be vectors in $\mathbb{R}^3$.*

1 ***Addition of vectors:***

$$\vec{a} + \vec{b} = \langle a_1 + b_1, a_2 + b_2, a_3 + b_3 \rangle$$

2 ***Scalar multiple of a vector:***

$$c\vec{a} = c\langle a_1, a_2, a_3 \rangle = \langle ca_1, ca_2, ca_3 \rangle$$

3 The ***zero vector 0****:*

$$\vec{0} = \langle 0, 0, 0 \rangle$$

4 The ***negative -a*** *of a vector $\vec{a}$:*

$$-\vec{a} = -\langle a_1, a_2, a_3 \rangle = \langle -a_1, -a_2, -a_3 \rangle$$

5 The ***difference a-b****:*

$$\vec{a} - \vec{b} = \vec{a} + (-\vec{b}) = \langle a_1 - b_1, a_2 - b_2, a_3 - b_3 \rangle$$

6 The length of $\vec{a} = \langle a_1, a_2, a_3 \rangle$:

$$\|\vec{a}\| = \sqrt{a_1^2 + a_2^2 + a_3^2}$$

7 If $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ are any points, the vector $\vec{a}$ in $\mathbb{R}^3$ that corresponds to $P_1\vec{P}_2$ is:

$$\vec{a} = \langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle$$

Definition 1.37 The **standard basis vectors** in three dimensions are:

$$\vec{i} = \langle 1, 0, 0 \rangle, \quad \vec{j} = \langle 0, 1, 0 \rangle, \quad \vec{k} = \langle 0, 0, 1 \rangle$$

If $\vec{a} = \langle a_1, a_2, a_3 \rangle$, then it can be written as a linear combination of the standard unit basis vectors:

$$\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$$

◆

Example 1.38 If $\vec{a} = \vec{i} + 2\vec{j} - 3\vec{k}$ and $\vec{b} = 4\vec{i} + 7\vec{k}$, express the vector $2\vec{a} + 3\vec{b}$ in terms of $\vec{i}$, $\vec{j}$, and $\vec{k}$.

Solution. Substitute the given expressions for $\vec{a}$ and $\vec{b}$, distribute the scalars, and group the components:

$$\begin{aligned} 2\vec{a} + 3\vec{b} &= 2(\vec{i} + 2\vec{j} - 3\vec{k}) + 3(4\vec{i} + 0\vec{j} + 7\vec{k}) \\ &= (2\vec{i} + 4\vec{j} - 6\vec{k}) + (12\vec{i} + 0\vec{j} + 21\vec{k}) \\ &= (2 + 12)\vec{i} + (4 + 0)\vec{j} + (-6 + 21)\vec{k} \\ &= 14\vec{i} + 4\vec{j} + 15\vec{k} \end{aligned}$$

▲

Example 1.39 Find the unit vector in the direction of the vector $\vec{v} = 2\vec{i} - \vec{j} - 2\vec{k}$.

Solution. First, find the magnitude of the given vector $\vec{v}$:

$$\|\vec{v}\| = \sqrt{2^2 + (-1)^2 + (-2)^2} = \sqrt{4 + 1 + 4} = \sqrt{9} = 3$$

Next, multiply the reciprocal of the magnitude by the vector $\vec{v}$ to produce the unit vector $\vec{u}$:

$$\vec{u} = \frac{1}{3}(2\vec{i} - \vec{j} - 2\vec{k}) = \frac{2}{3}\vec{i} - \frac{1}{3}\vec{j} - \frac{2}{3}\vec{k}$$

▲

Example 1.40 A 100-lb weight hangs from two wires making angles of 50° and 32° with the horizontal. Find the tensions (forces) $\vec{T_1}$ and $\vec{T_2}$ in both wires and their magnitudes.

Solution. We express the tension vectors in terms of their horizontal and vertical components. Let $\vec{T_1}$ pull up and to the left, and $\vec{T_2}$ pull up and to the right:

$$\begin{aligned} \vec{T_1} &= -\|\vec{T_1}\| \cos(50°)\vec{i} + \|\vec{T_1}\| \sin(50°)\vec{j} \\ \vec{T_2} &= \|\vec{T_2}\| \cos(32°)\vec{i} + \|\vec{T_2}\| \sin(32°)\vec{j} \end{aligned}$$

The downward force due to the weight is $\vec{w} = -100\vec{j}$. For the system to achieve static equilibrium, the resultant sum of the tensions must exactly counterbalance the weight:

$$\vec{T_1} + \vec{T_2} = -\vec{w} = 100\vec{j}$$

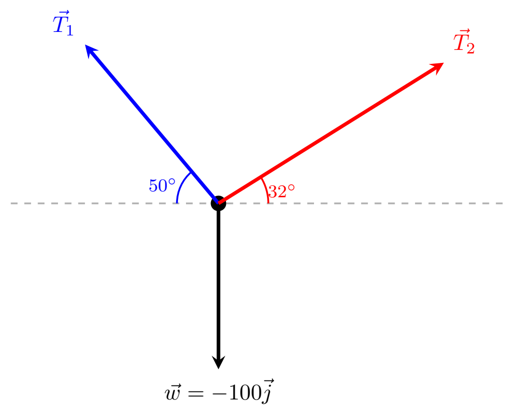


Figure 1.41 Force vector diagram for the hanging weight system.

Equating the horizontal ($\vec{i}$) and vertical ($\vec{j}$) components gives a system of linear equations:

$$-\|\vec{T}_1\|\cos(50^\circ) + \|\vec{T}_2\|\cos(32^\circ) = 0$$
$$\|\vec{T}_1\|\sin(50^\circ) + \|\vec{T}_2\|\sin(32^\circ) = 100$$

From the first equation, we find $\|\vec{T}_2\| = \|\vec{T}_1\|\frac{\cos(50^\circ)}{\cos(32^\circ)}$. Substituting this relationship into the second equation yields:

$$\|\vec{T}_1\|\sin(50^\circ) + \|\vec{T}_1\|\frac{\cos(50^\circ)\sin(32^\circ)}{\cos(32^\circ)} = 100$$

$$\|\vec{T}_1\|\left(\sin(50^\circ) + \cos(50^\circ)\tan(32^\circ)\right) = 100$$

Solving for the magnitudes yields:

- $\|\vec{T}_1\| \approx 85.64$ lb
- $\|\vec{T}_2\| = 85.64 \cdot \dfrac{\cos(50^\circ)}{\cos(32^\circ)} \approx 64.91$ lb

Substituting these scalar magnitudes back into our component formulas provides the final tension vector models:

$$\vec{T}_1 \approx -55.05\vec{i} + 65.60\vec{j}$$
$$\vec{T}_2 \approx 55.05\vec{i} + 34.40\vec{j}$$

▲

1.3 The Dot Product

Suppose a constant **force** is represented by a vector $\vec{F} = \vec{PR}$ pointing in some direction other than the direction of motion. If this force moves an object from point P to point Q, then the **displacement vector** is $\vec{D} = \vec{PQ}$.

The **work** done by $\vec{F}$ is defined as the magnitude of the displacement $\vec{D}$, multiplied by the magnitude of the applied force in the direction of the motion. Therefore, the work done by $\vec{F}$ is defined to be:

$$W = \|\vec{D}\|(\|\vec{F}\|\cos(\theta)) = \|\vec{D}\|\|\vec{F}\|\cos(\theta)$$

where θ is the angle between the force and displacement vectors.

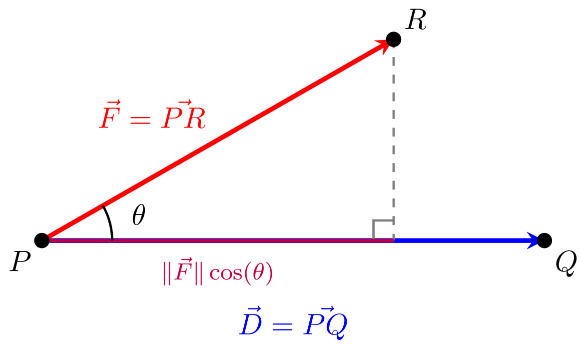


Figure 1.42 The geometric projection of a force vector $\vec{F}$ along a displacement vector $\vec{D}$.

Definition 1.43 The **dot product** of two vectors $\vec{a}$ and $\vec{b}$ is the number:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta)$$

where θ is the angle between $\vec{a}$ and $\vec{b}$, such that $0 \leq \theta \leq \pi$. Thus, θ is the smallest angle between the vectors when they are drawn with the same initial point. If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, we define $\vec{a} \cdot \vec{b} = 0$. ◆

The dot product is a real number, that is, a scalar. For this reason, the dot product is sometimes called the **scalar product**.

Example 1.44 If the vectors $\vec{a}$ and $\vec{b}$ have lengths 4 and 6, and the angle between them is $\pi/3$, find $\vec{a} \cdot \vec{b}$.

Solution. Substitute the given magnitudes and the angle into the geometric definition of the dot product:

$$\begin{aligned}\vec{a} \cdot \vec{b} &= \|\vec{a}\|\|\vec{b}\|\cos(\theta) \\ &= (4)(6)\cos\left(\frac{\pi}{3}\right) \\ &= 24 \cdot \left(\frac{1}{2}\right) \\ &= 12\end{aligned}$$

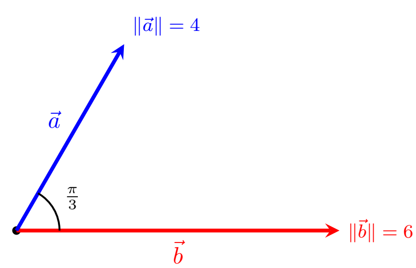


Figure 1.45 Two vectors separated by an angle of $\pi/3$.

▲

Example 1.46 A crate is hauled 8 m up a ramp under a constant force of 200 N applied at an angle of 25° to the ramp. Find the work done.

Solution. The work done by a constant force is given by the formula $W = \|\vec{F}\|\|\vec{D}\|\cos(\theta)$, which corresponds exactly to the dot product of the force vector and the displacement vector $\vec{F} \cdot \vec{D}$.

Given $\|\vec{F}\| = 200$ N, $\|\vec{D}\| = 8$ m, and $\theta = 25°$:

$$\begin{aligned} W &= (200)(8)\cos(25°) \\ &= 1600\cos(25°) \\ &\approx 1450.09 \text{ J} \end{aligned}$$

The work done in hauling the crate is approximately 1450.09 Joules (or Newton-meters).

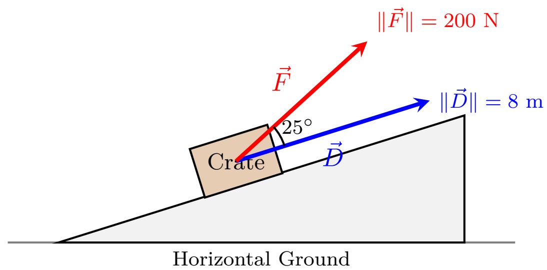


Figure 1.47 Force vector acting at an angle to the displacement along a ramp.

▲

Corollary 1.48 *Two nonzero vectors $\vec{a}$ and $\vec{b}$ are called **perpendicular** or **orthogonal** if the angle between them is $\theta = \pi/2$. In this case:*

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\pi/2) = 0$$

Conversely, if $\vec{a} \cdot \vec{b} = 0$ then $\cos(\theta) = 0$, which implies $\theta = \pi/2$. Therefore, two vectors $\vec{a}$ and $\vec{b}$ are orthogonal if and only if $\vec{a} \cdot \vec{b} = 0$.

Proof. By definition, the geometric dot product is given by $\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta)$, where $0 \leq \theta \leq \pi$.

For the forward direction, assume $\vec{a}$ and $\vec{b}$ are orthogonal. By definition, the angle between them is $\theta = \pi/2$. Substituting this value yields:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|\cos\left(\frac{\pi}{2}\right) = \|\vec{a}\|\|\vec{b}\|(0) = 0$$

For the reverse direction, assume $\vec{a} \cdot \vec{b} = 0$ for two nonzero vectors. Since the vectors are nonzero, their magnitudes are strictly positive ($\|\vec{a}\| \neq 0$ and $\|\vec{b}\| \neq 0$). We can divide both sides of the definition by these magnitudes:

$$0 = \|\vec{a}\|\|\vec{b}\|\cos(\theta) \implies \cos(\theta) = 0$$

Within the restricted interval $0 \leq \theta \leq \pi$, the equation $\cos(\theta) = 0$ has a unique solution, $\theta = \pi/2$. Thus, the vectors are orthogonal. ■

Corollary 1.49 *The sign of the dot product $\vec{a} \cdot \vec{b}$ provides geometric information about the relative directions of the vectors:*

1. *It is positive if $\vec{a}$ and $\vec{b}$ point in the same general direction ($0 \leq \theta < \pi/2$).*
2. *It is 0 if they are perpendicular ($\theta = \pi/2$).*
3. *It is negative if $\vec{a}$ and $\vec{b}$ point in generally opposite directions ($\pi/2 < \theta \leq \pi$).*
4. *If $\vec{a}$ and $\vec{b}$ point in exactly the same direction, then $\theta = 0$, which means*

$\cos(\theta) = 1$ *and:*

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\|$$

5 *If $\vec{a}$ and $\vec{b}$ point in exactly the opposite direction, then $\theta = \pi$, which means* $\cos(\theta) = -1$ *and:*

$$\vec{a} \cdot \vec{b} = -\|\vec{a}\|\|\vec{b}\|$$

Proof. Because the magnitudes $\|\vec{a}\|$ and $\|\vec{b}\|$ of two nonzero vectors are always strictly positive, the sign of the product $\|\vec{a}\|\|\vec{b}\|\cos(\theta)$ is completely determined by the sign of the trigonometric term $\cos(\theta)$ on the interval $[0, \pi]$:

1 When $0 \le \theta < \pi/2$ (an acute angle), $\cos(\theta) > 0$, rendering the dot product positive.

2 When $\theta = \pi/2$ (a right angle), $\cos(\theta) = 0$, rendering the dot product zero.

3 When $\pi/2 < \theta \le \pi$ (an obtuse angle), $\cos(\theta) < 0$, rendering the dot product negative.

4 If they point in the exact same direction, $\theta = 0$. Since $\cos(0) = 1$, the expression simplifies directly to $\|\vec{a}\|\|\vec{b}\|$.

5 If they point in the exact opposite direction, $\theta = \pi$. Since $\cos(\pi) = -1$, the expression simplifies directly to $-\|\vec{a}\|\|\vec{b}\|$.

■

The Dot Product in Component Form

Suppose $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ are given in component forms. If we apply the Law of Cosines to the triangle formed by these vectors, we obtain:

$$\|\vec{a} - \vec{b}\|^2 = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\|\vec{a}\|\|\vec{b}\|\cos(\theta) = \|\vec{a}\|^2 + \|\vec{b}\|^2 - 2\vec{a} \cdot \vec{b}$$

Fact 1.50 Algebraic Dot Product Formula. *The dot product of $\vec{a} = \langle a_1, a_2, a_3 \rangle$ and $\vec{b} = \langle b_1, b_2, b_3 \rangle$ is:*

$$\vec{a} \cdot \vec{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$$

Example 1.51 Find the dot product $\vec{a} \cdot \vec{b}$ for each of the following cases:

1 $\vec{a} = \langle 2, -3, 4 \rangle$, $\vec{b} = \langle 6, 2, -\frac{1}{2} \rangle$

2 $\vec{a} = 4\vec{j} - 3\vec{k}$, $\vec{b} = 2\vec{i} + 4\vec{j} + 6\vec{k}$

3 $\|\vec{a}\| = 6$, $\|\vec{b}\| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is $2\pi/3$.

Solution.

1 Multiply corresponding components and add them together:

$$\vec{a} \cdot \vec{b} = (2)(6) + (-3)(2) + (4)\left(-\frac{1}{2}\right) = 12 - 6 - 2 = 4$$

2 First write $\vec{a}$ in full component form as $\langle 0, 4, -3 \rangle$ and $\vec{b}$ as $\langle 2, 4, 6 \rangle$:

$$\vec{a} \cdot \vec{b} = (0)(2) + (4)(4) + (-3)(6) = 0 + 16 - 18 = -2$$

3 Since components are unknown but the angle is provided, use the geometric definition:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta) = (6)(5) \cos\left(\frac{2\pi}{3}\right) = 30\left(-\frac{1}{2}\right) = -15$$

▲

Example 1.52 Show that $2\vec{i} + 2\vec{j} - \vec{k}$ is perpendicular to $5\vec{i} - 4\vec{j} + 2\vec{k}$.

Solution. Two vectors are perpendicular (orthogonal) if and only if their dot product equals zero. Let $\vec{a} = \langle 2, 2, -1\rangle$ and $\vec{b} = \langle 5, -4, 2\rangle$:

$$\vec{a} \cdot \vec{b} = (2)(5) + (2)(-4) + (-1)(2) = 10 - 8 - 2 = 0$$

Because the dot product is exactly 0, the vectors are perpendicular. ▲

Example 1.53 Find the angle between the vectors $\vec{a} = \langle 2, 2, -1\rangle$ and $\vec{b} = \langle 5, -3, 2\rangle$.

Solution. We use the alternate formula for the angle between vectors: $\cos(\theta) = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\|\vec{b}\|}$.

First compute the dot product and individual vector magnitudes:

$$\vec{a} \cdot \vec{b} = (2)(5) + (2)(-3) + (-1)(2) = 10 - 6 - 2 = 2$$

$$\|\vec{a}\| = \sqrt{2^2 + 2^2 + (-1)^2} = \sqrt{4 + 4 + 1} = \sqrt{9} = 3$$
$$\|\vec{b}\| = \sqrt{5^2 + (-3)^2 + 2^2} = \sqrt{25 + 9 + 4} = \sqrt{38}$$

Substitute these calculations back into the cosine expression:

$$\cos(\theta) = \frac{2}{3\sqrt{38}} \implies \theta = \arccos\left(\frac{2}{3\sqrt{38}}\right) \approx 1.46 \text{ radians (or } 83.8^\circ)$$

▲

Proposition 1.54 Properties of the Dot Product. *If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors in $\mathbb{R}^3$ and c is a scalar, then:*

1 $\vec{a} \cdot \vec{a} = \|\vec{a}\|^2$

2 $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$

3 $\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$

4 $(c\vec{a}) \cdot \vec{b} = c(\vec{a} \cdot \vec{b}) = \vec{a} \cdot (c\vec{b})$

5 $\vec{0} \cdot \vec{a} = \vec{a} \cdot \vec{0} = 0$

Proof. Let $\vec{a} = \langle a_1, a_2, a_3\rangle$, $\vec{b} = \langle b_1, b_2, b_3\rangle$, and $\vec{c} = \langle c_1, c_2, c_3\rangle$.

1 By the component formula for the dot product:

$$\vec{a} \cdot \vec{a} = a_1a_1 + a_2a_2 + a_3a_3 = a_1^2 + a_2^2 + a_3^2 = (\|\vec{a}\|)^2 = \|\vec{a}\|^2$$

2 Since multiplication of real numbers is commutative ($a_ib_i = b_ia_i$):

$$\vec{a} \cdot \vec{b} = a_1b_1 + a_2b_2 + a_3b_3 = b_1a_1 + b_2a_2 + b_3a_3 = \vec{b} \cdot \vec{a}$$

3 Using the distributive property of real numbers:

$$\vec{a} \cdot (\vec{b} + \vec{c}) = \vec{a} \cdot \langle b_1 + c_1, b_2 + c_2, b_3 + c_3\rangle$$

$$= a_1(b_1 + c_1) + a_2(b_2 + c_2) + a_3(b_3 + c_3)$$
$$= (a_1b_1 + a_1c_1) + (a_2b_2 + a_2c_2) + (a_3b_3 + a_3c_3)$$
$$= (a_1b_1 + a_2b_2 + a_3b_3) + (a_1c_1 + a_2c_2 + a_3c_3) = \vec{a} \cdot \vec{b} + \vec{a} \cdot \vec{c}$$

4 By factoring the real scalar c from the algebraic terms:

$$(c\vec{a}) \cdot \vec{b} = (ca_1)b_1 + (ca_2)b_2 + (ca_3)b_3 = c(a_1b_1 + a_2b_2 + a_3b_3) = c(\vec{a} \cdot \vec{b})$$

A symmetric argument shows this is also equal to $\vec{a} \cdot (c\vec{b})$.

5 Since multiplying any real number by zero equals zero:

$$\vec{0} \cdot \vec{a} = (0)(a_1) + (0)(a_2) + (0)(a_3) = 0 + 0 + 0 = 0$$

■

Theorem 1.55 Cauchy-Schwarz Inequality. *For any two vectors $\vec{a}$ and $\vec{b}$:*

$$|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$$

Proof. If either $\vec{a} = \vec{0}$ or $\vec{b} = \vec{0}$, then both sides of the inequality are exactly 0, making the statement trivially true.

Assume both vectors are nonzero. We can express the dot product using its geometric definition:

$$\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta)$$

Taking the absolute value of both sides yields:

$$|\vec{a} \cdot \vec{b}| = |\|\vec{a}\|\|\vec{b}\| \cos(\theta)| = \|\vec{a}\|\|\vec{b}\||\cos(\theta)|$$

Since the cosine function is naturally bounded by the range of real values $-1 \leq \cos(\theta) \leq 1$, its absolute value parameter satisfies $|\cos(\theta)| \leq 1$. Substituting this boundary restriction into our equation produces the target proof expression:

$$|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|(1) \implies |\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$$

■

Theorem 1.56 Triangle Inequality. *For any two vectors $\vec{a}$ and $\vec{b}$:*

$$\|\vec{a} + \vec{b}\| \leq \|\vec{a}\| + \|\vec{b}\|$$

Proof. Because both sides of the target inequality are non-negative real lengths, we can square the left-hand side and evaluate it using dot product properties:

$$\begin{aligned} \|\vec{a} + \vec{b}\|^2 &= (\vec{a} + \vec{b}) \cdot (\vec{a} + \vec{b}) \\ &= \vec{a} \cdot \vec{a} + \vec{a} \cdot \vec{b} + \vec{b} \cdot \vec{a} + \vec{b} \cdot \vec{b} \\ &= \|\vec{a}\|^2 + 2(\vec{a} \cdot \vec{b}) + \|\vec{b}\|^2 \end{aligned}$$

Since a real number is always less than or equal to its absolute value ($x \leq |x|$), we can state that $\vec{a} \cdot \vec{b} \leq |\vec{a} \cdot \vec{b}|$. Applying the Cauchy-Schwarz Inequality ($|\vec{a} \cdot \vec{b}| \leq \|\vec{a}\|\|\vec{b}\|$) gives:

$$\begin{aligned} \|\vec{a} + \vec{b}\|^2 &\leq \|\vec{a}\|^2 + 2|\vec{a} \cdot \vec{b}| + \|\vec{b}\|^2 \\ &\leq \|\vec{a}\|^2 + 2\|\vec{a}\|\|\vec{b}\| + \|\vec{b}\|^2 \end{aligned}$$

The right side is a perfect square trinomial, which factors into scalar lengths:

$$\|\vec{a}+\vec{b}\|^2 \leq (\|\vec{a}\|+\|\vec{b}\|)^2$$

Taking the positive square root of both sides completes the proof:

$$\|\vec{a}+\vec{b}\| \leq \|\vec{a}\|+\|\vec{b}\|$$

■

Projections

Let $\vec{PQ}$ and $\vec{PR}$ be representations of two vectors $\vec{a}$ and $\vec{b}$ sharing the same initial point P. If S is the foot of the perpendicular dropped from R to the line containing $\vec{PQ}$, then the vector represented by $\vec{PS}$ is called the **vector projection** of $\vec{b}$ onto $\vec{a}$ and is denoted by $\operatorname{proj}_{\vec{a}} \vec{b}$ (think of this as the shadow cast by $\vec{b}$ onto $\vec{a}$).

The **scalar projection** of $\vec{b}$ onto $\vec{a}$ (also called the **component of b along a**) is the real number $\|\vec{b}\|\cos(\theta)$, where θ is the angle between $\vec{a}$ and $\vec{b}$. This value is denoted by $\operatorname{comp}_{\vec{a}} \vec{b}$. The equation:

$$\vec{a}\cdot\vec{b} = \|\vec{a}\|\|\vec{b}\|\cos(\theta) = \|\vec{a}\|(\|\vec{b}\|\cos(\theta))$$

shows that the dot product of $\vec{a}$ and $\vec{b}$ can be interpreted as the length of $\vec{a}$ multiplied by the scalar projection of $\vec{b}$ onto $\vec{a}$. Since:

$$\|\vec{b}\|\cos(\theta) = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|} = \frac{\vec{a}}{\|\vec{a}\|}\cdot\vec{b}$$

Fact 1.57 Projection Formulas. *Let $\vec{a}$ and $\vec{b}$ be vectors with $\vec{a} \neq \vec{0}$.*

- ***Scalar projection of $\vec{b}$ onto $\vec{a}$:***

$$\operatorname{comp}_{\vec{a}} \vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|}$$

- ***Vector projection of $\vec{b}$ onto $\vec{a}$:***

$$\operatorname{proj}_{\vec{a}} \vec{b} = \left(\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|}\right)\frac{\vec{a}}{\|\vec{a}\|} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|^2}\vec{a}$$

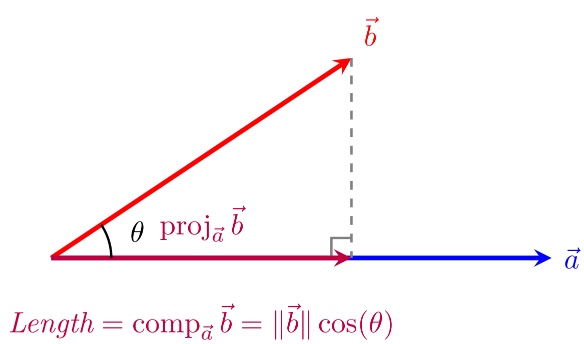


Figure 1.58 *The geometric relationship between vector projection and scalar projection.*

Example 1.59 Find the scalar projection and vector projection of $\vec{b} = \langle 1, 1, 2\rangle$ onto $\vec{a} = \langle -2, 3, 1\rangle$.

Solution. First, compute the core values required by the projection formulas ($\vec{a}\cdot\vec{b}$ and $\|\vec{a}\|^2$):

$$\vec{a}\cdot\vec{b} = (-2)(1) + (3)(1) + (1)(2) = -2 + 3 + 2 = 3$$

$$\|\vec{a}\|^2 = (-2)^2 + 3^2 + 1^2 = 4 + 9 + 1 = 14 \implies \|\vec{a}\| = \sqrt{14}$$

Now apply these numbers to the scalar projection formula:

$$\operatorname{comp}_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|} = \frac{3}{\sqrt{14}}$$

Next, apply the values to the vector projection formula to scale the base direction vector $\vec{a}$:

$$\operatorname{proj}_{\vec{a}}\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|^2}\vec{a} = \frac{3}{14}\langle -2, 3, 1\rangle = \left\langle -\frac{3}{7}, \frac{9}{14}, \frac{3}{14}\right\rangle$$

▲

Checkpoint 1.60 Find the direction angle between the vector $\vec{a} = \langle 1, 2, 3\rangle$ and the x-axis.

Solution. The direction angle relative to the positive x-axis matches the angle between vector $\vec{a}$ and the standard basis unit vector $\vec{i} = \langle 1, 0, 0\rangle$:

$$\vec{a}\cdot\vec{i} = (1)(1) + (2)(0) + (3)(0) = 1$$

$$\|\vec{a}\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$$
$$\|\vec{i}\| = 1$$

Now compute the angle tracking parameter α:

$$\cos(\alpha) = \frac{\vec{a}\cdot\vec{i}}{\|\vec{a}\|\|\vec{i}\|} = \frac{1}{\sqrt{14}} \implies \alpha = \arccos\left(\frac{1}{\sqrt{14}}\right) \approx 74.5^\circ$$

Checkpoint 1.61 A force is given by a vector $\vec{F} = 3\vec{i} + 4\vec{j} + 5\vec{k}$ and moves a particle from the point $P(2, 1, 0)$ to the point $Q(4, 6, 2)$. Find the work done.

Solution. First, determine the displacement vector $\vec{D}$ from P to Q using terminal minus initial coordinates:

$$\vec{D} = \vec{PQ} = \langle 4 - 2, 6 - 1, 2 - 0\rangle = \langle 2, 5, 2\rangle$$

Work is computed as the dot product of the force vector $\vec{F} = \langle 3, 4, 5\rangle$ and this displacement vector $\vec{D}$:

$$W = \vec{F}\cdot\vec{D} = (3)(2) + (4)(5) + (5)(2) = 6 + 20 + 10 = 36$$

The total work done is 36 units.

Checkpoint 1.62 A cart weighing 100 lb is pushed up an incline that makes an angle of 30° with the horizontal. Find the work done against gravity in pushing the cart a distance of 80 feet.

Solution. The work done against gravity depends entirely on the vertical displacement of the cart. Alternatively, using our standard projection model, the gravity force vector points straight down with magnitude 100 lb: $\vec{F}_g = -100\vec{j}$.

If we align our coordinate frame with the horizontal ground, a displacement of 80 feet along a 30° incline yields:

$$\vec{D} = 80\cos(30°)\vec{i} + 80\sin(30°)\vec{j} = 80\left(\frac{\sqrt{3}}{2}\right)\vec{i} + 80\left(\frac{1}{2}\right)\vec{j} = 40\sqrt{3}\vec{i} + 40\vec{j}$$

The work done by gravity is $\vec{F}_g \cdot \vec{D} = (0)(40\sqrt{3}) + (-100)(40) = -4000$ ft-lb. The work done **against** gravity is the positive opposite of this value:

$$W = 4000 \text{ ft-lb}$$

Checkpoint 1.63 A wagon is pulled a distance of 100 m along a horizontal path by a constant force of 70 N. The handle of the wagon is held at an angle of 35° above the horizontal. Find the work done by the force.

Solution. Work is defined as the magnitude of the displacement vector multiplied by the scalar projection of the force vector along the direction of motion:

$$W = \|\vec{D}\|\left(\text{comp}_{\vec{D}}\,\vec{F}\right) = \|\vec{D}\|\left(\|\vec{F}\|\cos(\theta)\right)$$

Substitute the given values ($\|\vec{D}\| = 100$ m, $\|\vec{F}\| = 70$ N, and $\theta = 35°$) into the work expression:

$$\begin{aligned} W &= (100)(70)\cos(35°) \\ &= 7000\cos(35°) \\ &\approx 5734.06 \text{ J} \end{aligned}$$

The total work done by the force pulling the wagon is approximately 5734.06 Joules.

1.4 The Cross Product (The vector product)

The cross product of two vectors in three-dimensional space is a vector that is perpendicular to both of the original vectors. It is defined as:

$$\vec{a} \times \vec{b} = \|\vec{a}\|\|\vec{b}\|\sin(\theta)\vec{n}$$

where θ is the angle between the vectors and $\vec{n}$ is the unit vector perpendicular to both $\vec{a}$ and $\vec{b}$.

Definition 1.64 If $\vec{a}$ and $\vec{b}$ are nonzero three-dimensional vectors, the **cross product** of $\vec{a}$ and $\vec{b}$ is the vector:

$$\vec{a} \times \vec{b} = (\|\vec{a}\|\|\vec{b}\|\sin(\theta))\vec{n}$$

where θ is the angle between $\vec{a}$ and $\vec{b}$ such that $0 \leq \theta \leq \pi$, and $\vec{n}$ is a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$. The direction of $\vec{n}$ is determined by the **right-hand rule**: if the fingers of your right hand curl through the angle θ from $\vec{a}$ to $\vec{b}$, then your thumb points in the direction of $\vec{n}$.

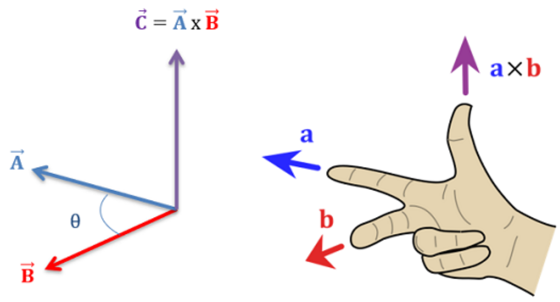


Figure 1.65 The right-hand rule and mutual vector orthogonality.

◆

Corollary 1.66 *The vector $\vec{a} \times \vec{b}$ points in the same direction as $\vec{n}$ because it is a scalar multiple of $\vec{n}$ with a non-negative scalar coefficient. Consequently, $\vec{a} \times \vec{b}$ is orthogonal to both $\vec{a}$ and $\vec{b}$. This is equivalent to stating:*

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = 0 \quad \textit{and} \quad (\vec{a} \times \vec{b}) \cdot \vec{b} = 0$$

Proof. By definition, the cross product is a scalar multiple of the vector $\vec{n}$, meaning $\vec{a} \times \vec{b} = c\vec{n}$ where $c = \|\vec{a}\|\|\vec{b}\| \sin(\theta)$. Because $\vec{n}$ is explicitly defined as a unit vector perpendicular to both $\vec{a}$ and $\vec{b}$, its dot products with them must evaluate to zero:

$$\vec{n} \cdot \vec{a} = 0 \quad \text{and} \quad \vec{n} \cdot \vec{b} = 0$$

Using the scalar distribution properties of the dot product, we can substitute $c\vec{n}$ back into the orthogonality checks:

$$(\vec{a} \times \vec{b}) \cdot \vec{a} = (c\vec{n}) \cdot \vec{a} = c(\vec{n} \cdot \vec{a}) = c(0) = 0$$

$$(\vec{a} \times \vec{b}) \cdot \vec{b} = (c\vec{n}) \cdot \vec{b} = c(\vec{n} \cdot \vec{b}) = c(0) = 0$$

This confirms that the resulting vector product remains perfectly perpendicular to both original structural components. ■

Corollary 1.67 *Two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if the angle between them is either 0 or π. Therefore, two nonzero vectors $\vec{a}$ and $\vec{b}$ are parallel if and only if:*

$$\vec{a} \times \vec{b} = \vec{0}$$

Proof. The magnitude of the cross product vector is given by $\|\vec{a} \times \vec{b}\| = \|\vec{a}\|\|\vec{b}\| \sin(\theta)$. For any vector, its overall value equals the zero vector ($\vec{0}$) if and only if its absolute magnitude length scales to exactly 0.

Because we are evaluating two strictly nonzero vectors, their isolated scalar magnitudes are positive ($\|\vec{a}\| \neq 0$ and $\|\vec{b}\| \neq 0$). Therefore, the product evaluates to zero if and only if the trigonometric term vanishes:

$$\|\vec{a} \times \vec{b}\| = 0 \iff \sin(\theta) = 0$$

On the restricted tracking interval $0 \leq \theta \leq \pi$, the condition $\sin(\theta) = 0$ is satisfied if and only if $\theta = 0$ or $\theta = \pi$. Geometrically, an angle of 0 means the vectors point in the exact same direction, and an angle of π means they point in exactly opposite directions. In either case, the vectors are parallel. ■

Example 1.68 Find $\vec{i} \times \vec{j}$ and $\vec{j} \times \vec{i}$.

Solution. We use the geometric definition of the cross product, $\vec{a} \times \vec{b} = (\|\vec{a}\|\|\vec{b}\| \sin(\theta))\vec{n}$:

- For $\vec{i} \times \vec{j}$, both vectors are unit vectors, so $\|\vec{i}\| = 1$ and $\|\vec{j}\| = 1$. The standard basis vectors are mutually perpendicular, meaning the angle between them is $\theta = \pi/2$. By the right-hand rule, curling your fingers from the positive x-axis ($\vec{i}$) to the positive y-axis ($\vec{j}$) forces your thumb to point along the positive z-axis ($\vec{k}$):
$$\vec{i} \times \vec{j} = (1 \cdot 1 \cdot \sin(\pi/2))\vec{k} = (1 \cdot 1 \cdot 1)\vec{k} = \vec{k}$$
- For $\vec{j} \times \vec{i}$, reversing the order of operations reverses the orientation of your hand gesture. The thumb now points down along the negative z-axis ($-\vec{k}$):
$$\vec{j} \times \vec{i} = (1 \cdot 1 \cdot \sin(\pi/2))(-\vec{k}) = -\vec{k}$$

▲

Remark 1.69 The cross product of any vector with itself is always the zero vector, because the angle between identical direction paths is $\theta = 0$, and $\sin(0) = 0$:
$$\vec{i} \times \vec{i} = \vec{0}, \quad \vec{j} \times \vec{j} = \vec{0}, \quad \vec{k} \times \vec{k} = \vec{0}$$

The Cross Product in Component Form

Fact 1.70 Algebraic Cross Product Formula. *The cross product of $\vec{a} = \langle a_1, a_2, a_3\rangle$ and $\vec{b} = \langle b_1, b_2, b_3\rangle$ is:*
$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1\rangle$$

The expression for $\vec{a} \times \vec{b}$ is easier to remember in determinant form. The cross product of $\vec{a} = a_1\vec{i} + a_2\vec{j} + a_3\vec{k}$ and $\vec{b} = b_1\vec{i} + b_2\vec{j} + b_3\vec{k}$ can be expanded along its top row as:
$$\vec{a} \times \vec{b} = \begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} \vec{i} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \vec{j} + \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix} \vec{k}$$

We compact this row-expansion memory tool by writing it as a formal 3×3 matrix determinant:
$$\vec{a} \times \vec{b} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}$$

Proposition 1.71 Properties of the Cross Product. *If $\vec{a}$, $\vec{b}$, and $\vec{c}$ are vectors in $\mathbb{R}^3$ and c is a scalar, then:*

1 $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$ *(Anti-commutativity)*

2 $(c\vec{a}) \times \vec{b} = c(\vec{a} \times \vec{b}) = \vec{a} \times (c\vec{b})$

3 $\vec{a} \times (\vec{b} + \vec{c}) = \vec{a} \times \vec{b} + \vec{a} \times \vec{c}$ *(Left distributivity)*

4 $(\vec{a} + \vec{b}) \times \vec{c} = \vec{a} \times \vec{c} + \vec{b} \times \vec{c}$ *(Right distributivity)*

Proof. Let $\vec{a} = \langle a_1, a_2, a_3\rangle$, $\vec{b} = \langle b_1, b_2, b_3\rangle$, and $\vec{c} = \langle c_1, c_2, c_3\rangle$. By algebraic definition, the cross product component vector expansion is:
$$\vec{a} \times \vec{b} = \langle a_2b_3 - a_3b_2,\ a_3b_1 - a_1b_3,\ a_1b_2 - a_2b_1\rangle$$

1 Expanding the reverse cross product $\vec{b} \times \vec{a}$ yields:

$$\vec{b} \times \vec{a} = \langle b_2a_3 - b_3a_2,\, b_3a_1 - b_1a_3,\, b_1a_2 - b_2a_1\rangle$$

Factoring out a negative -1 from each scalar entry directly flips the interior subtraction ordering:

$$\begin{aligned}\vec{b} \times \vec{a} &= \langle -(a_2b_3 - a_3b_2),\, -(a_3b_1 - a_1b_3),\, -(a_1b_2 - a_2b_1)\rangle \\ &= -\langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1\rangle = -(\vec{a} \times \vec{b})\end{aligned}$$

Multiplying both sides by -1 satisfies the anti-commutative condition $\vec{a} \times \vec{b} = -(\vec{b} \times \vec{a})$.

2 Multiplying vector $\vec{a}$ by a scalar c updates our definition to $c\vec{a} = \langle ca_1, ca_2, ca_3\rangle$. Substituting this scalar into the expansion gives:

$$\begin{aligned}(c\vec{a}) \times \vec{b} &= \langle (ca_2)b_3 - (ca_3)b_2,\, (ca_3)b_1 - (ca_1)b_3,\, (ca_1)b_2 - (ca_2)b_1\rangle \\ &= \langle c(a_2b_3 - a_3b_2),\, c(a_3b_1 - a_1b_3),\, c(a_1b_2 - a_2b_1)\rangle \\ &= c\langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1\rangle = c(\vec{a} \times \vec{b})\end{aligned}$$

A symmetric factoring loop shows this expression is identically equal to $\vec{a} \times (c\vec{b})$.

3 First, compile the vector addition term: $\vec{b} + \vec{c} = \langle b_1 + c_1, b_2 + c_2, b_3 + c_3\rangle$. Crossing $\vec{a}$ with this combined vector yields:

$$\begin{aligned}\vec{a} \times (\vec{b} + \vec{c}) &= \langle a_2(b_3 + c_3) - a_3(b_2 + c_2),\, a_3(b_1 + c_1) - a_1(b_3 + c_3),\, a_1(b_2 + c_2) - a_2(b_1 + c_1)\rangle \\ &= \langle a_2b_3 + a_2c_3 - a_3b_2 - a_3c_2,\, a_3b_1 + a_3c_1 - a_1b_3 - a_1c_3,\, a_1b_2 + a_1c_2 - a_2b_1 - a_2c_1\rangle\end{aligned}$$

Regrouping the entries splits the single vector back into a clean linear addition step:

$$\begin{aligned}&= \langle (a_2b_3 - a_3b_2) + (a_2c_3 - a_3c_2),\, (a_3b_1 - a_1b_3) + (a_3c_1 - a_1c_3),\, (a_1b_2 - a_2b_1) + (a_1c_2 - a_2c_1)\rangle \\ &= \langle a_2b_3 - a_3b_2,\, a_3b_1 - a_1b_3,\, a_1b_2 - a_2b_1\rangle + \langle a_2c_3 - a_3c_2,\, a_3c_1 - a_1c_3,\, a_1c_2 - a_2c_1\rangle \\ &= \vec{a} \times \vec{b} + \vec{a} \times \vec{c}\end{aligned}$$

4 Property 4 can be proven directly by using the anti-commutative law established in step 1, combined with the left distributive property from step 3:

$$\begin{aligned}(\vec{a} + \vec{b}) \times \vec{c} &= -(\vec{c} \times (\vec{a} + \vec{b})) \\ &= -(\vec{c} \times \vec{a} + \vec{c} \times \vec{b}) \\ &= -(\vec{c} \times \vec{a}) - (\vec{c} \times \vec{b}) \\ &= (\vec{a} \times \vec{b}) + (\vec{b} \times \vec{c})\end{aligned}$$

This completes the verification of all four arithmetic identities.

■

If $\vec{a}$ and $\vec{b}$ are represented by directed line segments with the same initial point, they determine a parallelogram. This parallelogram has a base of length $\|\vec{a}\|$ and an altitude height of $\|\vec{b}\|\sin(\theta)$. Therefore, the total area A is:

$$A = \|\vec{a}\|\|\vec{b}\|\sin(\theta) = \|\vec{a} \times \vec{b}\|$$

Remark 1.72 The length of the cross product vector $\|\vec{a} \times \vec{b}\|$ is equal to the area of the parallelogram determined by the vectors $\vec{a}$ and $\vec{b}$.

Example 1.73 Find the cross product $\vec{a} \times \vec{b}$ for each of the following cases:

1 $\vec{a} = \langle 2, -3, 4 \rangle$, $\vec{b} = \langle 6, 2, -\frac{1}{2} \rangle$

2 $\vec{a} = 4\vec{j} - 3\vec{k}$, $\vec{b} = 2\vec{i} + 4\vec{j} + 6\vec{k}$

3 $\|\vec{a}\| = 6$, $\|\vec{b}\| = 5$, and the angle between $\vec{a}$ and $\vec{b}$ is $2\pi/3$.

Solution.

1 Set up and evaluate the 3×3 determinant:

$$\begin{aligned}
\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -3 & 4 \\ 6 & 2 & -\frac{1}{2} \end{vmatrix} \\
&= \begin{vmatrix} -3 & 4 \\ 2 & -\frac{1}{2} \end{vmatrix} \vec{i} - \begin{vmatrix} 2 & 4 \\ 6 & -\frac{1}{2} \end{vmatrix} \vec{j} + \begin{vmatrix} 2 & -3 \\ 6 & 2 \end{vmatrix} \vec{k} \\
&= \left(\frac{3}{2} - 8 \right) \vec{i} - (-1 - 24)\vec{j} + (4 - (-18))\vec{k} \\
&= -\frac{13}{2}\vec{i} + 25\vec{j} + 22\vec{k} = \left\langle -\frac{13}{2}, 25, 22 \right\rangle
\end{aligned}$$

2 Write $\vec{a}$ in full component form as $\langle 0, 4, -3 \rangle$:

$$\begin{aligned}
\vec{a} \times \vec{b} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 0 & 4 & -3 \\ 2 & 4 & 6 \end{vmatrix} \\
&= (24 - (-12))\vec{i} - (0 - (-6))\vec{j} + (0 - 8)\vec{k} \\
&= 36\vec{i} - 6\vec{j} - 8\vec{k} = \langle 36, -6, -8 \rangle
\end{aligned}$$

3 Because individual components are missing but the angle is given, we find the magnitude using the geometric formula and label the unit normal directional vector $\vec{n}$:

$$\|\vec{a} \times \vec{b}\| = \|\vec{a}\|\|\vec{b}\| \sin(\theta) = (6)(5) \sin\left(\frac{2\pi}{3}\right) = 30\left(\frac{\sqrt{3}}{2}\right) = 15\sqrt{3}$$

Thus, the cross product vector is $15\sqrt{3}\vec{n}$.

▲

Example 1.74 Find a vector perpendicular to the plane that passes through the points $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

Solution. First, construct two independent vectors lying inside the plane using the given vertices:

$$\vec{PQ} = \langle -2 - 1, 5 - 4, -1 - 6 \rangle = \langle -3, 1, -7 \rangle$$

$$\vec{PR} = \langle 1 - 1, -1 - 4, 1 - 6 \rangle = \langle 0, -5, -5 \rangle$$

A vector perpendicular to the plane must be orthogonal to both paths. Compute their cross product:

$$\begin{aligned}\vec{PQ} \times \vec{PR} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ -3 & 1 & -7 \\ 0 & -5 & -5 \end{vmatrix} \\ &= (-5-35)\vec{i} - (15-0)\vec{j} + (15-0)\vec{k} \\ &= -40\vec{i} - 15\vec{j} + 15\vec{k} = \langle -40, -15, 15 \rangle\end{aligned}$$

Any non-zero scalar multiple of this vector (such as $\langle 8, 3, -3 \rangle$ obtained by dividing by -5) is also perpendicular to the plane. ▲

Example 1.75 Find the area of the triangle with vertices $P(1, 4, 6)$, $Q(-2, 5, -1)$, and $R(1, -1, 1)$.

Solution. The area of a triangle is exactly half the area of the parallelogram spanned by vectors sharing vertex P. From the previous example, we know $\vec{PQ} \times \vec{PR} = \langle -40, -15, 15 \rangle$.

Compute the magnitude length of this cross product vector:

$$\begin{aligned}\|\vec{PQ} \times \vec{PR}\| &= \sqrt{(-40)^2 + (-15)^2 + 15^2} \\ &= \sqrt{1600 + 225 + 225} = \sqrt{2050} = 5\sqrt{82}\end{aligned}$$

Therefore, the area of the triangle is:

$$\text{Area} = \frac{1}{2}\|\vec{PQ} \times \vec{PR}\| = \frac{5\sqrt{82}}{2} \approx 22.64$$

▲

Definition 1.76 The product $\vec{a} \cdot (\vec{b} \times \vec{c})$ is called the **scalar triple product** of the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. ◆

The area of the base parallelogram is $A = \|\vec{b} \times \vec{c}\|$. If θ is the angle between vector $\vec{a}$ and the cross product vector $\vec{b} \times \vec{c}$, then the slanted vertical height h of the surrounding three-dimensional parallelepiped is $h = \|\vec{a}\||\cos(\theta)|$.

Fact 1.77 Volume of a Parallelepiped. *The volume of the parallelepiped determined by the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ is the absolute magnitude of their scalar triple product:*

$$V = |\vec{a} \cdot (\vec{b} \times \vec{c})| = |(\vec{a} \times \vec{b}) \cdot \vec{c}|$$

We can compute this product directly using a single matrix determinant:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$$

Example 1.78 Use the scalar triple product to show that the vectors $\vec{a} = \langle 1, 4, -7 \rangle$, $\vec{b} = \langle 2, -1, 4 \rangle$, and $\vec{c} = \langle 0, -9, 18 \rangle$ are coplanar, meaning they lie in the same plane.

Solution. Three vectors are coplanar if and only if the volume of the parallelepiped they span is exactly zero ($V = 0$). We check this condition by computing their scalar triple product determinant:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} 1 & 4 & -7 \\ 2 & -1 & 4 \\ 0 & -9 & 18 \end{vmatrix}$$

$$
\begin{aligned}
&= 1\begin{vmatrix} -1 & 4 \\ -9 & 18 \end{vmatrix} - 4\begin{vmatrix} 2 & 4 \\ 0 & 18 \end{vmatrix} + (-7)\begin{vmatrix} 2 & -1 \\ 0 & -9 \end{vmatrix} \\
&= 1(-18 - (-36)) - 4(36 - 0) - 7(-18 - 0) \\
&= 1(18) - 4(36) - 7(-18) \\
&= 18 - 144 + 126 = 0
\end{aligned}
$$

Because the scalar triple product is exactly 0, the bounded volume vanishes. This proves that the three vectors are coplanar. ▲

Definition 1.79 The product $\vec{a} \times (\vec{b} \times \vec{c})$ is called the **vector triple product** of the vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$. It can be expanded and computed using dot products via the BAC-CAB identity:

$$\vec{a} \times (\vec{b} \times \vec{c}) = (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c}$$

◆

Example 1.80 Given the vectors $\vec{a} = \vec{i} - \vec{j} + 2\vec{k}$, $\vec{b} = 2\vec{i} + \vec{k}$, and $\vec{c} = \vec{j} - 3\vec{k}$, evaluate the vector triple product $\vec{a} \times (\vec{b} \times \vec{c})$ using the expansion identity.

Solution. First, express the vectors cleanly in their numerical component forms:

$$\vec{a} = \langle 1, -1, 2 \rangle, \quad \vec{b} = \langle 2, 0, 1 \rangle, \quad \vec{c} = \langle 0, 1, -3 \rangle$$

Next, calculate the two scalar dot products required by the identity ($\vec{a} \cdot \vec{c}$ and $\vec{a} \cdot \vec{b}$):

$$
\begin{aligned}
\vec{a} \cdot \vec{c} &= (1)(0) + (-1)(1) + (2)(-3) = 0 - 1 - 6 = -7 \\
\vec{a} \cdot \vec{b} &= (1)(2) + (-1)(0) + (2)(1) = 2 + 0 + 2 = 4
\end{aligned}
$$

Substitute these scalar values back into the vector triple product identity formula:

$$
\begin{aligned}
\vec{a} \times (\vec{b} \times \vec{c}) &= (\vec{a} \cdot \vec{c})\vec{b} - (\vec{a} \cdot \vec{b})\vec{c} \\
&= -7\langle 2, 0, 1 \rangle - 4\langle 0, 1, -3 \rangle \\
&= \langle -14, 0, -7 \rangle - \langle 0, 4, -12 \rangle \\
&= \langle -14 - 0,\ 0 - 4,\ -7 - (-12) \rangle = \langle -14, -4, 5 \rangle
\end{aligned}
$$

In standard basis notation, the final vector is $-14\vec{i} - 4\vec{j} + 5\vec{k}$. ▲

1.5 Lines and Planes

In this section, we will discuss the equations of lines and planes in three-dimensional space. We will explore how to represent lines and planes using vector equations, parametric equations, and Cartesian equations. Additionally, we will examine the relationships between lines and planes, including conditions for parallelism and intersection.

If a vector $\vec{v}$ gives the direction of a line L and is written in component form as $\vec{v} = \langle a, b, c \rangle$, then any scalar multiple of that direction vector can be expressed as $t\vec{v} = \langle ta, tb, tc \rangle$ for any scalar parameter t.

Letting $\vec{r} = \langle x, y, z \rangle$ represent the position vector of an arbitrary point on the line, and $\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$ represent the position vector of a specific known point on the line, we can state the **vector equation of a line**:

$$\vec{r} = \vec{r}_0 + t\vec{v} \implies \langle x, y, z \rangle = \langle x_0 + ta,\ y_0 + tb,\ z_0 + tc \rangle$$

Definition 1.81 Two vectors are equal if and only if their corresponding components are equal. Equating the components of the vector equation of a line yields:

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

where $t \in \mathbb{R}$. These equations are called the **parametric equations** of the line L passing through the point $P_0(x_0, y_0, z_0)$ and parallel to the vector $\vec{v} = \langle a, b, c\rangle$. Each value of the scalar parameter t represents a unique point (x, y, z) on L. ◆

Example 1.82 Find a vector equation and parametric equations for the line that passes through the point $(5, 1, 3)$ and is parallel to the vector $\vec{i} + 4\vec{j} - 2\vec{k}$. Then, find two other points on this line.

Solution. We are given a point on the line corresponding to the initial position vector $\vec{r}_0 = \langle 5, 1, 3\rangle$ and a parallel direction vector $\vec{v} = \vec{i} + 4\vec{j} - 2\vec{k} = \langle 1, 4, -2\rangle$.

The **vector equation** is given by:

$$\vec{r} = \vec{r}_0 + t\vec{v} \implies \langle x, y, z\rangle = \langle 5, 1, 3\rangle + t\langle 1, 4, -2\rangle$$

By separating this expression into independent components, we find the **parametric equations**:

$$x = 5 + t, \quad y = 1 + 4t, \quad z = 3 - 2t$$

To find two other points on the line, we substitute any two non-zero values for the parameter t:

- Letting $t = 1$ yields: $x = 5+1 = 6$, $y = 1+4(1) = 5$, and $z = 3-2(1) = 1$. This gives the point $(6, 5, 1)$.
- Letting $t = -1$ yields: $x = 5 - 1 = 4$, $y = 1 + 4(-1) = -3$, and $z = 3 - 2(-1) = 5$. This gives the point $(4, -3, 5)$.

▲

Definition 1.83 If a vector $\vec{v} = \langle a, b, c\rangle$ is used to describe the direction of a line L, then the numbers a, b, and c are called the **direction numbers** of L. Since any vector parallel to $\vec{v}$ can also be used, any three numbers proportional to a, b, and c can serve as a valid set of direction numbers for L.

If we solve each of the parametric equations for the parameter t and equate the results (assuming none of the direction numbers a, b, or c are zero), we eliminate t to obtain:

$$\frac{x - x_0}{a} = \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

These expressions are called the **symmetric equations** of L. ◆

For instance, if a direction number is zero, say $a = 0$, we eliminate the parameter t for the remaining components and write the equations of L as:

$$x = x_0, \quad \frac{y - y_0}{b} = \frac{z - z_0}{c}$$

This means geometrically that the line L lies entirely within the vertical plane $x = x_0$.

Example 1.84 Find parametric equations and symmetric equations for the line that passes through the points $(2, 4, -3)$ and $(3, -1, 1)$. At what point does this line intersect the xy-plane?

Solution. First, find a direction vector $\vec{v}$ for the line by subtracting the coordinates of the first point from the second point:

$$\vec{v} = \langle 3-2,\ -1-4,\ 1-(-3)\rangle = \langle 1, -5, 4\rangle$$

Using the first point $(2, 4, -3)$ as our base position vector $\vec{r}_0 = \langle 2, 4, -3\rangle$, we find the **parametric equations**:

$$x = 2 + t, \quad y = 4 - 5t, \quad z = -3 + 4t$$

By solving each equation for the parameter t, we obtain the matching **symmetric equations**:

$$x - 2 = \frac{y-4}{-5} = \frac{z+3}{4}$$

The line intersects the xy-plane when its height component is zero ($z = 0$). Substitute $z = 0$ into the symmetric equations to solve for the coordinate positions:

$$x - 2 = \frac{0+3}{4} \implies x = 2 + \frac{3}{4} = \frac{11}{4}$$

$$\frac{y-4}{-5} = \frac{3}{4} \implies y - 4 = -\frac{15}{4} \implies y = 4 - \frac{15}{4} = \frac{1}{4}$$

Therefore, the point of intersection with the xy-plane is $\left(\frac{11}{4}, \frac{1}{4}, 0\right)$. ▲

Corollary 1.85 *A line segment starting from position vector $\vec{r}_0$ and ending at position vector $\vec{r}_1$ has a direction vector defined by $\vec{v} = \vec{r}_1 - \vec{r}_0$. Substituting this into the standard vector equation of a line $\vec{r} = \vec{r}_0 + t\vec{v}$ yields the **vector equation of a line segment**:*

$$\vec{r}(t) = (1-t)\vec{r}_0 + t\vec{r}_1, \quad 0 \le t \le 1$$

Example 1.86 Show that the lines L_1 and L_2 with the following parametric equations are **skew lines** (meaning they do not intersect and are not parallel):

$$\begin{aligned} L_1: &\quad x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t \\ L_2: &\quad x = 2s, \quad y = 3 + s, \quad z = -3 + 4s \end{aligned}$$

Solution. First, check if the lines are parallel by comparing their direction vectors:

$$\vec{v}_1 = \langle 1, 3, -1\rangle \quad \text{and} \quad \vec{v}_2 = \langle 2, 1, 4\rangle$$

Since the components are not proportional ($\frac{1}{2} \neq \frac{3}{1}$), the direction vectors are not scalar multiples of each other. Therefore, the lines are not parallel.

Next, test if the lines intersect by equating their corresponding coordinate expressions to see if a common parameter set (t, s) exists:

$$\begin{aligned} 1 + t &= 2s \quad \text{(Eq. 1)} \\ -2 + 3t &= 3 + s \quad \text{(Eq. 2)} \\ 4 - t &= -3 + 4s \quad \text{(Eq. 3)} \end{aligned}$$

From Equation 1, we get $t = 2s - 1$. Substitute this expression into Equation 2 to solve for s:

$$-2 + 3(2s-1) = 3 + s \implies -2 + 6s - 3 = 3 + s$$

$$5s = 8 \implies s = \frac{8}{5}$$

Using $s = \frac{8}{5}$, we find the value for t: $t = 2\left(\frac{8}{5}\right) - 1 = \frac{16}{5} - \frac{5}{5} = \frac{11}{5}$.

Now, substitute both parameters $s = \frac{8}{5}$ and $t = \frac{11}{5}$ into Equation 3 to see if they satisfy the vertical height component:

$$\text{Left Side: } 4 - t = 4 - \frac{11}{5} = \frac{20}{5} - \frac{11}{5} = \frac{9}{5}$$

$$\text{Right Side: } -3 + 4s = -3 + 4\left(\frac{8}{5}\right) = -\frac{15}{5} + \frac{32}{5} = \frac{17}{5}$$

Since $\frac{9}{5} \neq \frac{17}{5}$, the system of linear equations has no solution. This means the lines do not intersect. Because they are neither parallel nor intersecting, L_1 and L_2 are skew lines.

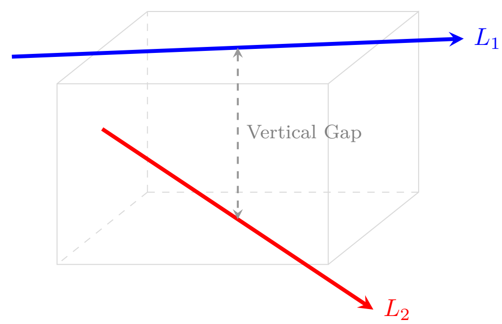


Figure 1.87 The skew lines L_1 and L_2 passing through different planes in space.

▲

Planes

A plane in space is uniquely determined by a fixed point $P_0(x_0, y_0, z_0)$ lying in the plane and a vector $\vec{n}$ that stands orthogonal to the plane surface. This perpendicular vector $\vec{n}$ is called a **normal vector**. Letting $P(x, y, z)$ be any arbitrary point in the plane, and denoting $\vec{r}_0$ and $\vec{r}$ as the standard position vectors of P_0 and P respectively, the displacement vector $\vec{r} - \vec{r}_0$ is represented by the geometric line segment $\vec{P_0P}$.

Since the normal vector $\vec{n}$ is orthogonal to every vector contained within the plane, it must be orthogonal to $\vec{r} - \vec{r}_0$. This orthogonality condition gives us the fundamental **vector equation of the plane**:

$$\vec{n} \cdot (\vec{r} - \vec{r}_0) = 0 \iff \vec{n} \cdot \vec{r} = \vec{n} \cdot \vec{r}_0$$

Theorem 1.88 Scalar and Linear Equations of a Plane. *An equation of the plane passing through $P_0(x_0, y_0, z_0)$ with normal vector $\vec{n} = \langle a, b, c\rangle$ is derived by evaluating the dot product component-wise:*

$$\begin{aligned} \langle a, b, c\rangle \cdot (\vec{r} - \vec{r}_0) &= 0 \\ \langle a, b, c\rangle \cdot \langle x - x_0, y - y_0, z - z_0\rangle &= 0 \\ a(x - x_0) + b(y - y_0) + c(z - z_0) &= 0 \end{aligned}$$

*By distributing the scalar coefficients and gathering the constants into a single value $d = -(ax_0 + by_0 + cz_0)$, the equation of the plane simplifies to the **linear equation in three variables**:*

$$ax + by + cz + d = 0$$

where the coefficients a, b, and c are not all zero.

Example 1.89 Find an equation of the plane through $(5, -2, 4)$ with normal vector $\vec{n} = \langle 1, 2, 3 \rangle$. Find its intercepts and describe how to sketch the plane.

Solution. Substitute the given coordinates and normal components directly into the scalar equation formula:

$$1(x-5) + 2(y-(-2)) + 3(z-4) = 0 \implies x - 5 + 2y + 4 + 3z - 12 = 0$$

Combining constant terms yields the final linear plane equation:

$$x + 2y + 3z = 13$$

To calculate the coordinate intercepts, set the other two variables to zero systematically:

- **x-intercept**: Set $y = 0, z = 0 \implies x = 13$. The point is $(13, 0, 0)$.
- **y-intercept**: Set $x = 0, z = 0 \implies 2y = 13 \implies y = \frac{13}{2}$. The point is $\left(0, \frac{13}{2}, 0\right)$.
- **z-intercept**: Set $x = 0, y = 0 \implies 3z = 13 \implies z = \frac{13}{2}$. The point is $\left(0, 0, \frac{13}{3}\right)$.

To sketch this plane within the first octant, plot these three intercept markers on the coordinate axes and connect them to construct a triangular boundary plane section. ▲

Example 1.90 Find an equation of the plane determined by the points $P(4, -3, 1)$, $Q(6, -4, 7)$, and $R(1, 2, 2)$.

Solution. First, construct two displacement vectors sharing vertex P that lie entirely within the plane:

$$\vec{PQ} = \langle 6-4, -4-(-3), 7-1 \rangle = \langle 2, -1, 6 \rangle$$

$$\vec{PR} = \langle 1-4, 2-(-3), 2-1 \rangle = \langle -3, 5, 1 \rangle$$

A normal vector $\vec{n}$ for the plane must stand perpendicular to both vector directions. We find it by computing their cross product:

$$\begin{aligned} \vec{n} = \vec{PQ} \times \vec{PR} &= \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & -1 & 6 \\ -3 & 5 & 1 \end{vmatrix} \\ &= (-1-30)\vec{i} - (2-(-18))\vec{j} + (10-3)\vec{k} \\ &= -31\vec{i} - 20\vec{j} + 7\vec{k} = \langle -31, -20, 7 \rangle \end{aligned}$$

Using point $P(4, -3, 1)$ and our new normal coordinates, assemble the scalar equation layout:

$$-31(x-4) - 20(y-(-3)) + 7(z-1) = 0 \implies -31x + 124 - 20y - 60 + 7z - 7 = 0$$

Simplifying constants yields the clean linear plane model:

$$-31x - 20y + 7z + 57 = 0 \quad \text{or} \quad 31x + 20y - 7z = 57$$

▲

Example 1.91 Find the point at which the line with parametric equations $x = 2 + 3t$, $y = -4t$, $z = 5 + t$ intersects the plane $4x + 5y - 2z = 18$.

Solution. Substitute the parametric component equations of the line directly into the independent variable slots of the linear plane:

$$4(2+3t)+5(-4t)-2(5+t)=18$$

Distribute coefficients and combine terms to isolate the scalar parameter t:

$$8+12t-20t-10-2t=18 \implies -10t-2=18 \implies -10t=20 \implies t=-2$$

Substitute $t=-2$ back into the original parametric expressions to recover the intersection point coordinates:

$$x=2+3(-2)=-4, \quad y=-4(-2)=8, \quad z=5+(-2)=3$$

Thus, the line intersects the plane at the specific point coordinate $(-4, 8, 3)$.

▲

Definition 1.92 Two distinct planes containing normal vectors $\vec{n}_1$ and $\vec{n}_2$ are:

1 **parallel** if their corresponding normal vectors are parallel ($\vec{n}_1 = c\vec{n}_2$ for some scalar c).

2 **orthogonal** if their corresponding normal vectors are orthogonal ($\vec{n}_1 \cdot \vec{n}_2 = 0$).

◆

Example 1.93 Find an equation of the plane through $P(5, -2, 4)$ that is parallel to the plane $3x+y-6z+8=0$.

Solution. Parallel planes share identical directional orientations, meaning we can adopt the normal vector of the reference plane directly. Reading the coefficients from $3x+y-6z+8=0$ reveals $\vec{n} = \langle 3, 1, -6\rangle$.

Combine this normal direction with the coordinates of point $P(5, -2, 4)$:

$$3(x-5)+1(y-(-2))-6(z-4)=0 \implies 3x-15+y+2-6z+24=0$$

Simplifying the constants provides the equation of the parallel plane:

$$3x+y-6z+11=0$$

▲

Example 1.94 Find the angle between the planes $x+y+z=1$ and $x-2y+3z=1$. Then, find symmetric equations for the line of intersection L of these two planes.

Solution. The angle θ between two planes matches the angle separating their normal vectors, $\vec{n}_1 = \langle 1,1,1\rangle$ and $\vec{n}_2 = \langle 1,-2,3\rangle$:

$$\vec{n}_1 \cdot \vec{n}_2 = (1)(1)+(1)(-2)+(1)(3)=2$$

$$\|\vec{n}_1\| = \sqrt{1^2+1^2+1^2} = \sqrt{3}, \quad \|\vec{n}_2\| = \sqrt{1^2+(-2)^2+3^2} = \sqrt{14}$$

$$\cos(\theta) = \frac{\vec{n}_1 \cdot \vec{n}_2}{\|\vec{n}_1\|\|\vec{n}_2\|} = \frac{2}{\sqrt{42}} \implies \theta = \arccos\left(\frac{2}{\sqrt{42}}\right) \approx 72.0^\circ \text{ (or 1.26 rad)}$$

To track the line of intersection L, we first find its direction vector $\vec{v}$, which must be orthogonal to both normal paths:

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 1 & 1 \\ 1 & -2 & 3 \end{vmatrix} = (1(3)-1(-2))\vec{i} - (1(3)-1(1))\vec{j} + (1(-2)-1(1))\vec{k} = 5\vec{i} - 2\vec{j} - 3\vec{k}$$

$$\vec{v} = \langle 5, -2, -3 \rangle$$

Next, we find a specific point on L by solving the system of plane equations simultaneously. We can set $z = 0$ to simplify:

$$x + y = 1 \quad \text{and} \quad x - 2y = 1$$

Solving this system yields $x = \frac{3}{2}$ and $y = -\frac{1}{2}$. Thus, the point $P\left(\frac{3}{2}, -\frac{1}{2}, 0\right)$ lies on the line of intersection.

Finally, we can write the parametric equations for the line L:

$$x = \frac{3}{2} + 5t, \quad y = -\frac{1}{2} - 2t, \quad z = -3t$$

And the symmetric equations are:

$$\frac{x - \frac{3}{2}}{5} = \frac{y + \frac{1}{2}}{-2} = \frac{z}{-3}$$

▲

Fact 1.95 Symmetric Forms and Intersecting Planes. *An arbitrary spatial point $P(x, y, z)$ lies on a line L if and only if its coordinates satisfy the standard* ***symmetric equations****:*

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

We can interpret this line structural model as the geometric intersection path generated by two distinct, non-parallel structural planes:

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} \quad \textit{and} \quad \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

Example 1.96 Find a symmetric form for the line passing through $P(3, 1, -2)$ and $Q(-2, 7, -4)$.

Solution. Calculate the baseline direction numbers by computing the displacement vector components from P to Q:

$$\vec{v} = \langle -2 - 3,\ 7 - 1,\ -4 - (-2) \rangle = \langle -5, 6, -2 \rangle$$

Using coordinates from vertex $P(3, 1, -2)$, populate the symmetric structural denominators:

$$\frac{x - 3}{-5} = \frac{y - 1}{6} = \frac{z + 2}{-2}$$

▲

Example 1.97 Find a formula for the minimum straight-line distance D mapping from an isolated spatial point $P_1(x_1, y_1, z_1)$ down to a linear plane surface defined by $ax + by + cz + d = 0$.

Solution. Let $P_0(x_0, y_0, z_0)$ be any arbitrary known point lying inside the given plane equation, which means it satisfies $ax_0 + by_0 + cz_0 + d = 0 \implies d = -ax_0 - by_0 - cz_0$. Construct a direction vector $\vec{b}$ corresponding to the directed line segment $\overrightarrow{P_0P_1}$:

$$\vec{b} = \langle x_1 - x_0,\ y_1 - y_0,\ z_1 - z_0 \rangle$$

The minimum distance D from point P_1 to the plane surface equals the absolute scalar projection length of this displacement vector $\vec{b}$ projected onto the plane's normal vector direction $\vec{n} = \langle a, b, c\rangle$:

$$\begin{aligned} D = |\operatorname{comp}_{\vec{n}} \vec{b}| &= \frac{|\vec{n} \cdot \vec{b}|}{\|\vec{n}\|} \\ &= \frac{|a(x_1 - x_0) + b(y_1 - y_0) + c(z_1 - z_0)|}{\sqrt{a^2 + b^2 + c^2}} \\ &= \frac{|ax_1 + by_1 + cz_1 - (ax_0 + by_0 + cz_0)|}{\sqrt{a^2 + b^2 + c^2}} \end{aligned}$$

Substituting the constant expression $d = -ax_0 - by_0 - cz_0$ into the numerator yields the standard point-plane distance formula:

$$D = \frac{|ax_1 + by_1 + cz_1 + d|}{\sqrt{a^2 + b^2 + c^2}}$$

▲

Checkpoint 1.98 Find the distance between the parallel planes $10x+2y-2z = 5$ and $5x + y - z = 1$.

Solution. To find the distance between parallel planes, pick an arbitrary test point on one plane and compute its distance to the other plane.

Let us find a point on the second plane $5x + y - z = 1$ by setting $x = 0$ and $z = 0$, which gives $y = 1$. This identifies our test point as $P_1(0, 1, 0)$.

Now, write the first plane in standard linear form: $10x + 2y - 2z - 5 = 0$. Apply the coordinates of $P_1(0, 1, 0)$ to the distance formula:

$$D = \frac{|10(0) + 2(1) - 2(0) - 5|}{\sqrt{10^2 + 2^2 + (-2)^2}} = \frac{|2 - 5|}{\sqrt{100 + 4 + 4}} = \frac{|-3|}{\sqrt{108}} = \frac{3}{6\sqrt{3}} = \frac{1}{2\sqrt{3}} = \frac{\sqrt{3}}{6}$$

The exact distance separating the two planes is $\frac{\sqrt{3}}{6}$.

Checkpoint 1.99 We showed that the lines L_1 and L_2 with the following parametric equations are skew lines:

$$\begin{aligned} L_1: &\quad x = 1 + t, \quad y = -2 + 3t, \quad z = 4 - t \\ L_2: &\quad x = 2s, \quad y = 3 + s, \quad z = -3 + 4s \end{aligned}$$

Find the distance between them.

Solution. The minimum distance between two skew lines equals the distance between two parallel planes that enclose the lines. The normal vector $\vec{n}$ for these planes must stand perpendicular to both line direction vectors: $\vec{v}_1 = \langle 1, 3, -1\rangle$ and $\vec{v}_2 = \langle 2, 1, 4\rangle$.

$$\vec{n} = \vec{v}_1 \times \vec{v}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 3 & -1 \\ 2 & 1 & 4 \end{vmatrix} = (12-(-1))\vec{i}-(4-(-2))\vec{j}+(1-6)\vec{k} = \langle 13, -6, -5\rangle$$

Find a point on line L_1 by setting $t = 0$, giving $P_1(1, -2, 4)$. Find a point on line L_2 by setting $s = 0$, giving $P_2(0, 3, -3)$.

Construct a plane passing through $P_2(0, 3, -3)$ using our calculated normal vector components:

$$13(x - 0) - 6(y - 3) - 5(z - (-3)) = 0 \implies 13x - 6y - 5z + 3 = 0$$

Finally, calculate the distance from point $P_1(1, -2, 4)$ on line L_1 to this new plane container:

$$D = \frac{|13(1) - 6(-2) - 5(4) + 3|}{\sqrt{13^2 + (-6)^2 + (-5)^2}} = \frac{|13 + 12 - 20 + 3|}{\sqrt{169 + 36 + 25}} = \frac{8}{\sqrt{230}}$$

The minimum distance separating the two skew lines is exactly $\frac{8}{\sqrt{230}}$.

Checkpoint 1.100 Find the distance from the point $(1, 1, 2)$ to the plane $3x + 2y + 6z = 6$.

Solution. First, rewrite the plane equation in standard linear form: $3x + 2y + 6z - 6 = 0$. This provides coefficients of $a = 3, b = 2, c = 6$, and $d = -6$.

Substitute these parameters along with the target point coordinates $(x_1, y_1, z_1) = (1, 1, 2)$ into the point-plane distance formula:

$$D = \frac{|3(1) + 2(1) + 6(2) - 6|}{\sqrt{3^2 + 2^2 + 6^2}} = \frac{|3 + 2 + 12 - 6|}{\sqrt{9 + 4 + 36}} = \frac{|11|}{\sqrt{49}} = \frac{11}{7}$$

The minimum straight-line distance from the point to the plane is exactly $\frac{11}{7}$.

1.6 Surfaces

In this section, we will explore the geometry of surfaces in three-dimensional space. We will learn how to represent surfaces using vector equations, parametric equations, and scalar equations. Additionally, we will discuss the relationships between surfaces and other geometric objects.

Definition 1.101 A **function** f **of two variables** is a rule that assigns to each ordered pair of real numbers (x, y) in a set D a unique real number denoted by $f(x, y)$. The set D is the **domain** of f and its **range** is the set of values that f takes on, that is:

$$\{f(x, y) \mid (x, y) \in D\}$$

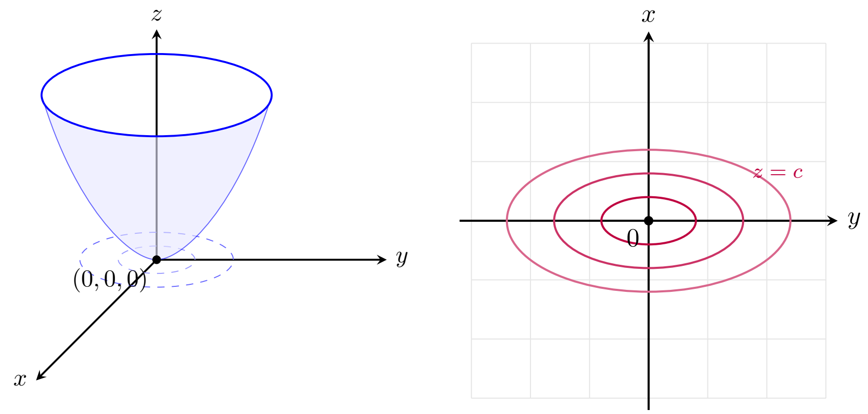
◆

We often write $z = f(x, y)$ to make explicit the value taken on by f at the general point (x, y). The domain is a subset of $\mathbb{R}^2$, the xy-plane.

Example 1.102 If $f(x, y) = 4x^2 + y^2$, determine its domain and range.

Solution. The expression $4x^2 + y^2$ is defined for all possible ordered pairs of real numbers (x, y). Therefore, the domain is $\mathbb{R}^2$, which represents the entire xy-plane.

Since the terms x^2 and y^2 are always non-negative for any real numbers x and y, their linear combination $4x^2 + y^2$ must also be greater than or equal to zero. The function achieves its absolute minimum value of 0 exactly at the origin $(0, 0)$. Thus, the range of f is the interval $[0, \infty)$ of all non-negative real numbers.



(a) The 3D surface (elliptic paraboloid).

(b) The domain and level curves in the xy-plane.

Figure 1.103 The 3D surface graph of $f(x, y) = 4x^2 + y^2$ and its corresponding domain and level curves in the xy-plane.

▲

Definition 1.104 If f is a function of two variables with domain D, then the **graph** of f is the set of all points (x, y, z) in $\mathbb{R}^3$ such that $z = f(x, y)$ and (x, y) is in D. ◆

Example 1.105 Sketch the graph of each of the following functions or equations in $\mathbb{R}^3$:

1 $f(x, y) = 6 - 3x - 2y$

2 An **elliptic cylinder**: $\frac{x^2}{4} + \frac{y^2}{9} = 1$

3 A **parabolic cylinder**: $f(x, y) = x^2$

4 An **ellipsoid**: $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$

Solution.

1 Setting $z = f(x, y)$ gives the linear equation $z = 6 - 3x - 2y \implies 3x + 2y + z = 6$. This represents a flat plane in space. We can find its coordinate axis intercepts to help sketch it:

- x-intercept: Set $y = 0, z = 0 \implies 3x = 6 \implies x = 2$. Point: $(2, 0, 0)$.
- y-intercept: Set $x = 0, z = 0 \implies 2y = 6 \implies y = 3$. Point: $(0, 3, 0)$.
- z-intercept: Set $x = 0, y = 0 \implies z = 6$. Point: $(0, 0, 6)$.

Plotting these three points and connecting them with a triangular frame yields the trace of the plane in the first octant.

2 The equation $\frac{x^2}{4} + \frac{y^2}{9} = 1$ restricts the relationship between x and y to an ellipse centered at the origin in the xy-plane, with major axis radius 3 along the y-axis and minor axis radius 2 along the x-axis. Because the variable z is completely missing from the equation, it is unconstrained and can take any real value. This projects the ellipse vertically to form an infinitely tall **elliptic cylinder** parallel to the z-axis.

3 Setting $z = f(x, y) = x^2$ provides a parabolic trace in the vertical xz-plane opening upwards from the origin. Since the variable y is absent, any vertical plane intersection cross-section matching $y = k$ produces an identical copy of this parabola shifted along the horizontal axis. Connecting these parallel parabolic traces generates a sheet-like **parabolic cylinder** extending along the direction of the y-axis.

4 The equation $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$ matches the standard form of an ellipsoid centered at the origin: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ with radii $a = 1$, $b = 3$, and $c = 2$. Its traces in all three coordinate planes are ellipses:

- xy-plane trace ($z = 0$): $x^2 + \frac{y^2}{9} = 1$ (an ellipse of radii 1 and 3).
- xz-plane trace ($y = 0$): $x^2 + \frac{z^2}{4} = 1$ (an ellipse of radii 1 and 2).
- yz-plane trace ($x = 0$): $\frac{y^2}{9} + \frac{z^2}{4} = 1$ (an ellipse of radii 3 and 2).

Sketching these three cross-sectional traces outlines a 3D egg-like oval surface shell.

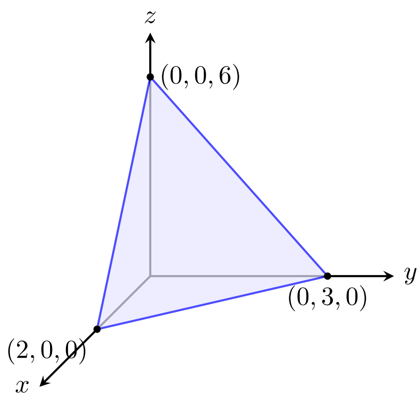


Figure 1.106 The plane $3x+2y+z = 6$ in the first octant.

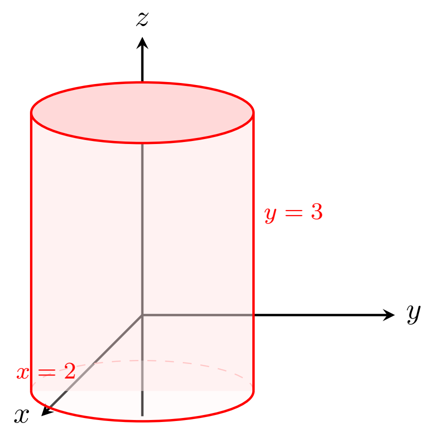


Figure 1.107 The elliptic cylinder $\frac{x^2}{4} + \frac{y^2}{9} = 1$.

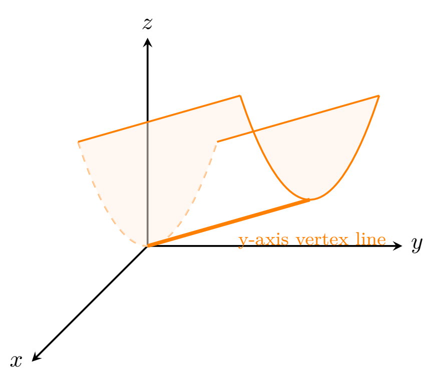


Figure 1.108 The parabolic cylinder $z = x^2$.

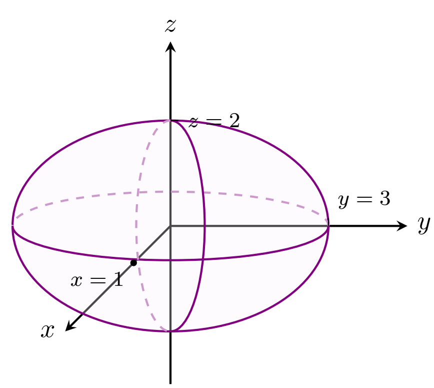


Figure 1.109 The ellipsoid $x^2 + \frac{y^2}{9} + \frac{z^2}{4} = 1$.

▲

If we keep x fixed by putting $x = k$ and letting y vary, the resulting function is a function of one variable, $z = f(k, y)$, whose graph is the curve that results when we intersect the surface $z = f(x, y)$ with the vertical plane $x = k$. This type of curve is called a **trace** of the surface. Standard quadric surfaces classified by their traces include:

- A **linear function**: $f(x, y) = ax + by + c$

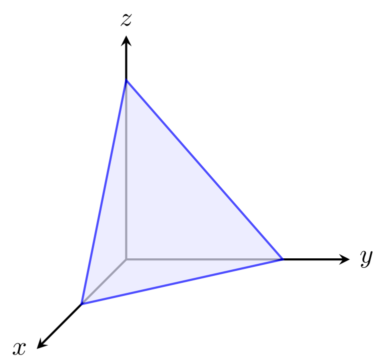


Figure 1.110 A flat plane representing a linear function.

- An **elliptic paraboloid**: $\frac{z}{c} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

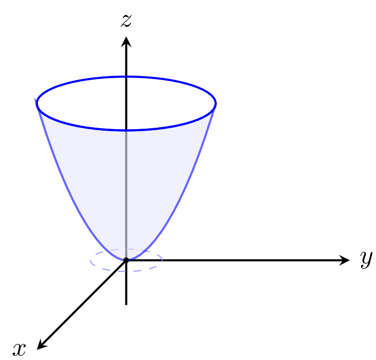


Figure 1.111 An elliptic paraboloid opening upward.

- A **hyperbolic paraboloid**: $\frac{z}{c} = \frac{x^2}{a^2} - \frac{y^2}{b^2}$

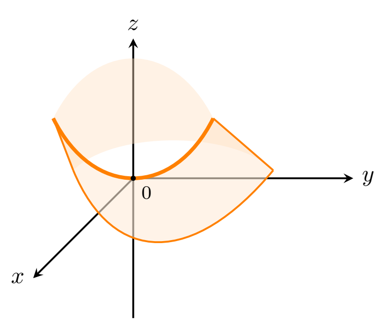


Figure 1.112 A hyperbolic paraboloid displaying a saddle point at the origin.

- An **ellipsoid**: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

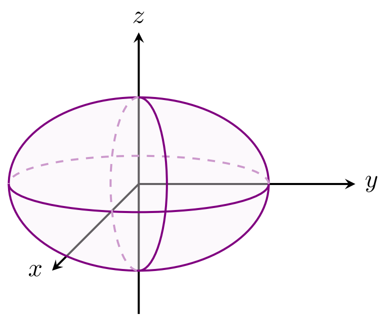


Figure 1.113 A symmetrical ellipsoid centered at the origin.

- A **hyperboloid of one sheet**: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

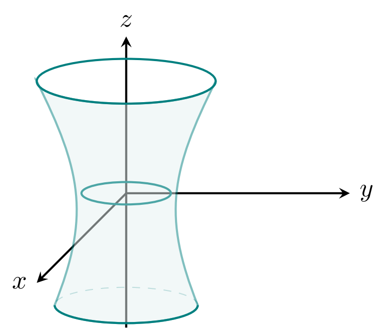


Figure 1.114 A hyperboloid of one sheet.

- A **hyperboloid of two sheets**: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

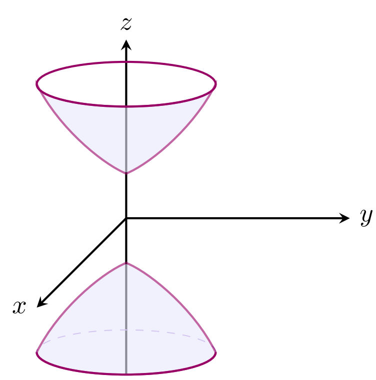


Figure 1.115 A hyperboloid of two sheets.

- An **elliptic cone**: $\frac{z^2}{c^2} = \frac{x^2}{a^2} + \frac{y^2}{b^2}$

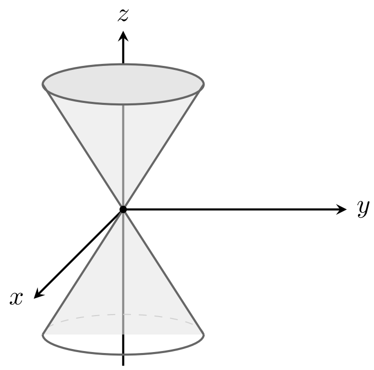


Figure 1.116 An elliptic cone.

Example 1.117 Classify the surface $x^2 + 2z^2 - 6x - y + 10 = 0$.

Solution. To classify the surface, we isolate the linear variable y and complete the square for the x terms:

$$
\begin{aligned}
y &= x^2 - 6x + 2z^2 + 10 \\
y &= (x^2 - 6x + 9) + 2z^2 + 10 - 9 \\
y - 1 &= (x-3)^2 + 2z^2
\end{aligned}
$$

We can rewrite this in standard form as:

$$y - 1 = \frac{(x-3)^2}{1} + \frac{z^2}{1/2}$$

Comparing this to our standard models, this equation represents an **elliptic paraboloid**. Its vertex is shifted to the coordinates $(3, 1, 0)$, and it opens along the positive direction of the y-axis.

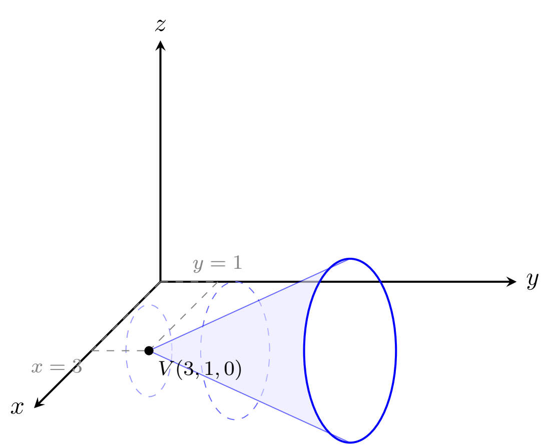


Figure 1.118 The shifted elliptic paraboloid $y - 1 = (x-3)^2 + 2z^2$.

▲

Chapter 2

Vector-valued Functions

Vector-valued functions are functions that take one or more variables and return a vector. They are used to describe curves and surfaces in three-dimensional space.

In this chapter, we will study the properties of vector-valued functions, including limits, continuity, and differentiability.

2.1 Vector-valued Functions and space curves

A vector-valued function is a function that takes one or more variables and returns a vector. It can be represented in the form:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle$$

where $x(t)$, $y(t)$, and $z(t)$ are scalar functions of the variable t.

Definition 2.1 A **vector-valued function** is a function whose domain is a set of real numbers and whose range is a set of vectors. Let D be a set of real numbers. Then the three components of the vector-valued function $\mathbf{r}(t)$ are uniquely determined:

$$\mathbf{r}(t) = \langle x(t), y(t), z(t) \rangle = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k}$$

where x, y, and z are scalar functions with the domain D. ◆

Example 2.2 If $\mathbf{r}(t) = \langle t^3, \ln(3-t), \sqrt{t} \rangle$, find the component functions.

Solution. By identifying the scalar functions multiplying each unit vector or filling each slot, the component functions are:

$$x(t) = t^3, \quad y(t) = \ln(3-t), \quad z(t) = \sqrt{t}$$

▲

Example 2.3 Let $\mathbf{r}(t) = \langle t+1, t^2-4, t^2 \rangle$ for $t \in \mathbb{R}$.

1. Sketch $\mathbf{r}(1)$ and $\mathbf{r}(3)$.
2. Find the vectors $\mathbf{r}(t)$ that are in a coordinate plane.

Solution.

1. The positions vectors are evaluated at the given parameter inputs:

$$\mathbf{r}(1) = \langle 1+1, 1^2-4, 1^2 \rangle = \langle 2, -3, 1 \rangle$$

$$\mathbf{r}(3) = \langle 3+1, 3^2-4, 3^2\rangle = \langle 4,5,9\rangle$$

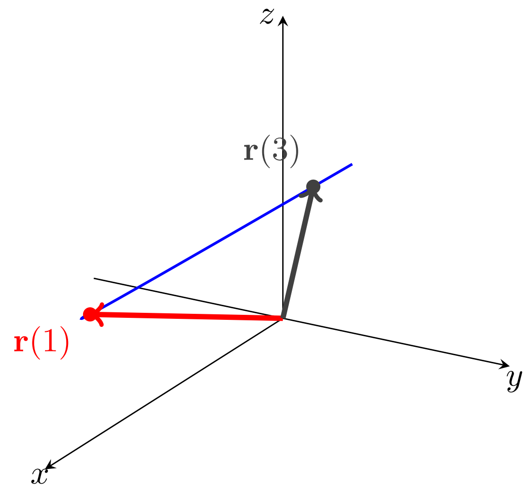


Figure 2.4 A 3D coordinate system showing the evaluation points on the path viewed at a 120° horizontal rotation.

2. We analyze each coordinate plane by setting the appropriate component to zero:

   (a) xy-plane : The **k**-component is zero.

   $$\begin{aligned} t^2 = 0 &\implies t = 0 \\ &\implies \mathbf{r}(0) = \langle 1, -4, 0\rangle \end{aligned}$$

   (b) yz-plane : The **i**-component is zero.

   $$\begin{aligned} t+1 = 0 &\implies t = -1 \\ &\implies \mathbf{r}(-1) = \langle 0, -3, 1\rangle \end{aligned}$$

   (c) xz-plane : The **j**-component is zero.

   $$\begin{aligned} t^2-4 = 0 &\implies t = 2 \text{ or } t = -2 \\ &\implies \mathbf{r}(2) = \langle 3, 0, 4\rangle \\ &\text{and } \mathbf{r}(-2) = \langle -1, 0, 4\rangle \end{aligned}$$

▲

Limits and Continuity

Definition 2.5 The **limit** of a vector function $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$ is defined by taking the limits of its component functions:

$$\lim_{t\to a}\mathbf{r}(t) = \langle \lim_{t\to a} x(t), \lim_{t\to a} y(t), \lim_{t\to a} z(t)\rangle$$

provided the limits of the component functions exist.

A vector function $\mathbf{r}(t)$ is **continuous at** a if:

$$\lim_{t\to a}\mathbf{r}(t) = \mathbf{r}(a)$$

Equivalently, $\mathbf{r}(t)$ is continuous at a if and only if its component functions x, y, and z are continuous at a. ◆

Example 2.6 Find $\lim_{t\to 0} \mathbf{r}(t)$, where:

$$\mathbf{r}(t) = \left\langle 1 + t^3, te^{-t}, \frac{\sin t}{t} \right\rangle$$

Solution. We evaluate the limit by computing the limit of each component function individually as $t \to 0$:

$$\lim_{t\to 0}(1 + t^3) = 1 + 0^3 = 1$$
$$\lim_{t\to 0}(te^{-t}) = 0 \cdot e^0 = 0$$
$$\lim_{t\to 0}\left(\frac{\sin t}{t}\right) = 1$$

Combining these results into a single vector gives:

$$\lim_{t\to 0} \mathbf{r}(t) = \langle 1, 0, 1 \rangle$$

▲

Suppose that f, g, and h are continuous on an interval I. Then the set C of all points (x, y, z) in space, where:

$$x = f(t), \quad y = g(t), \quad z = h(t)$$

and t varies in I, is called a **space curve**. The equations are called the **parametric equations of C** with a **parameter** t.

Example 2.7 Describe the curve defined by the vector function:

$$\mathbf{r}(t) = \langle 1 + t, 2 + 5t, -1 + 6t \rangle$$

Solution. The corresponding parametric equations are:

$$x = 1 + t, \quad y = 2 + 5t, \quad z = -1 + 6t$$

These are the parametric equations of a line passing through the point $(1, 2, -1)$ and parallel to the vector $\langle 1, 5, 6 \rangle$.

Alternatively, we can observe that the function can be written as:

$$\mathbf{r}(t) = \mathbf{r}_0 + t\mathbf{v}, \quad \text{where } \mathbf{r}_0 = \langle 1, 2, -1 \rangle \text{ and } \mathbf{v} = \langle 1, 5, 6 \rangle$$

which is the vector equation of a line. ▲

Example 2.8 Sketch the curve whose vector equation is:

$$\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + t\mathbf{k}$$

This curve is called a **helix**. This occurs in the model of DNA.

Solution. As the parameter t increases, the x- and y-components trace out a circle of radius 1 in the horizontal plane via $x = \cos t$ and $y = \sin t$. Concurrently, the z-component increases linearly since $z = t$. The combination of this circular base rotation and vertical motion generates a spiral path that climbs upward along the surface of a cylinder.

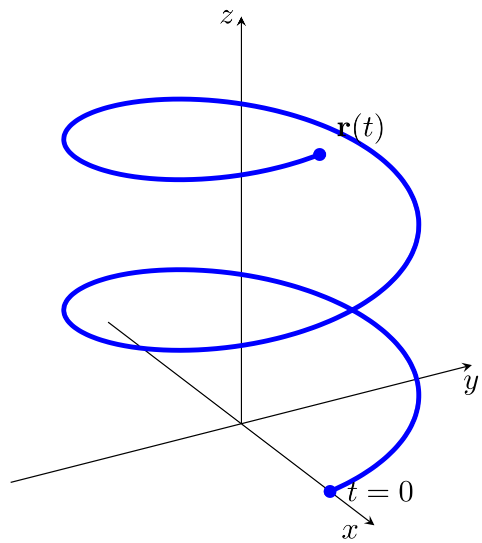


Figure 2.9 A 3D coordinate system tracing the path of a circular helix.

▲

Recall a vector equation for the line segment that joins the tip of the vector $\mathbf{r}_0$ to the tip of the vector $\mathbf{r}_1$:

$$\mathbf{r}(t) = (1-t)\mathbf{r}_0 + t\mathbf{r}_1 \qquad 0 \le t \le 1$$

Example 2.10 Find a vector equation and parametric equations for the line segment that joins the point $P(1, 3, -2)$ to the point $Q(2, -1, 3)$.

Solution. Using the points as components for the position vectors $\mathbf{r}_0 = \langle 1, 3, -2\rangle$ and $\mathbf{r}_1 = \langle 2, -1, 3\rangle$, the vector equation is:

$$\mathbf{r}(t) = \langle 1+t,\ 3-4t,\ -2+5t\rangle, \quad 0 \le t \le 1$$

The corresponding parametric equations are:

$$x = 1+t, \quad y = 3-4t, \quad z = -2+5t \qquad 0 \le t \le 1$$

▲

Example 2.11 Find a vector function that represents the curve of the intersection of the cylinder $x^2 + y^2 = 1$ and the plane $y + z = 2$.

Solution. The projection of the intersection curve C onto the xy-plane is the unit circle $x^2 + y^2 = 1$ with $z = 0$. We can parameterize this circular footprint as:

$$x = \cos t, \quad y = \sin t \qquad 0 \le t \le 2\pi$$

To find the corresponding z-coordinate along the path, substitute the expression for y into the plane equation $y + z = 2$:

$$z = 2 - y = 2 - \sin t$$

Thus, the parametric equations for C are $x = \cos t$, $y = \sin t$, and $z = 2 - \sin t$.

The corresponding vector function is:

$$\mathbf{r}(t) = \cos t\mathbf{i} + \sin t\mathbf{j} + (2 - \sin t)\mathbf{k} \qquad 0 \le t \le 2\pi$$

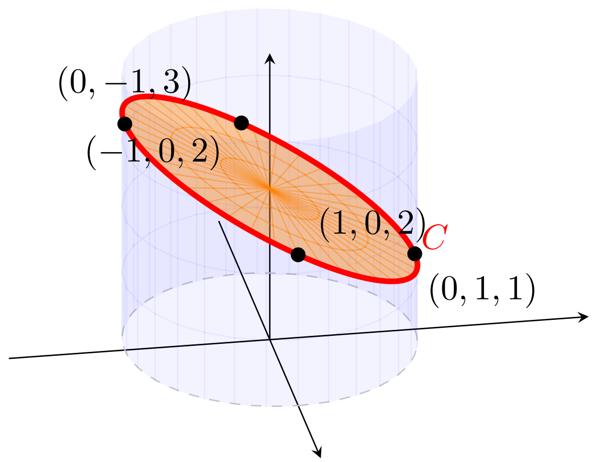


Figure 2.12 The colored intersection region viewed from an 80° horizontal rotation to separate tracking nodes.

▲

Example 2.13 Sketch the curve whose vector equation is:

$$\mathbf{r}(t) = t\mathbf{i} + t^2\mathbf{j} + t^3\mathbf{k} \quad t \ge 0$$

This curve is called a **twisted cubic**. You can view this curve as the intersection of the cylinders $y = x^2$ and $z = x^3$.

Solution. As t increases from 0, the path traces out a curve where the y-coordinate grows quadratically relative to x, and the z-coordinate grows cubically.

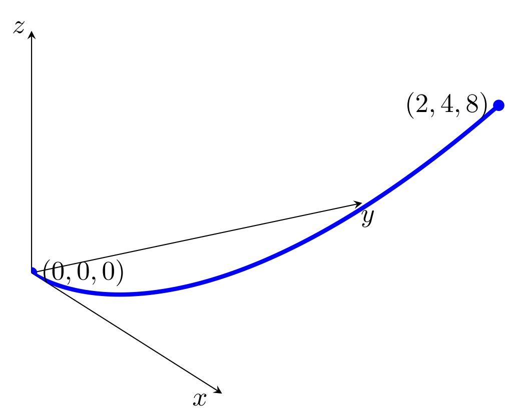


Figure 2.14 The parametric path of a twisted cubic curve for $t \ge 0$.

▲

Checkpoint 2.15 Let $\mathbf{r}(t) = 2t\mathbf{i} + (8 - 2t^2)\mathbf{j}$ for $-2 \le t \le 2$. Sketch the curve C defined by $\mathbf{r}(t)$ and indicate its orientation.

Solution. Eliminating the parameter yields $t = x/2$. Substituting this into the y-component gives the Cartesian equation:

$$y = 8 - 2\left(\frac{x}{2}\right)^2 = 8 - \frac{1}{2}x^2$$

This is a downward-opening parabola. As t goes from -2 to 2, the curve is traced from left to right starting at $(-4, 0)$, passing through the vertex at $(0, 8)$, and ending at $(4, 0)$.

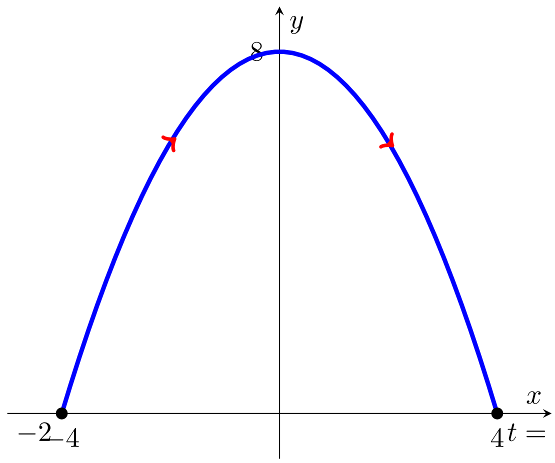


Figure 2.16 Downward parabola showing left-to-right orientation arrows.

Checkpoint 2.17 Let $\mathbf{r}(t) = \left(\frac{1}{2}t\cos t\right)\mathbf{i} + \left(\frac{1}{2}t\sin t\right)\mathbf{j}$ for $0 \le t \le 3\pi$. Sketch the curve C defined by $\mathbf{r}(t)$ and indicate its orientation.

Solution. In polar coordinates, this curve can be viewed as the spiral $r = \frac{1}{2}\theta$, where the parameter t acts as the angle θ. As t increases from 0 to 3π, the point winds counterclockwise outward from the origin, completing one and a half full revolutions.

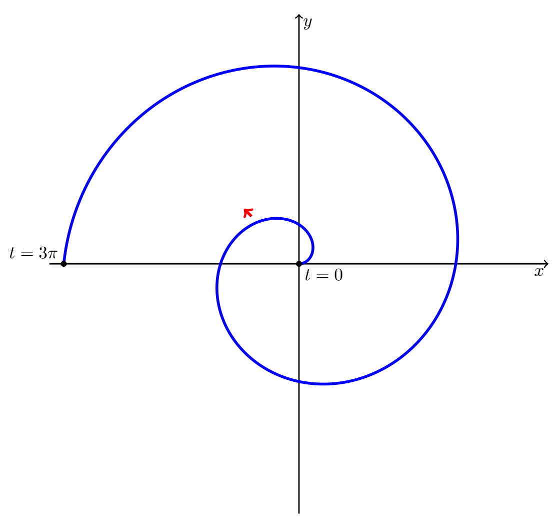


Figure 2.18 An Archimedean spiral winding outward counterclockwise with directional indicators.

2.2 Derivatives of Vector Functions

In this section, we will explore the concept of derivatives of vector functions. A vector function is a function that takes one or more variables and returns a vector. The derivative of a vector function provides information about how the vector changes with respect to its input variables.

We will cover the following topics:

- Definition of the derivative of a vector function
- Rules for differentiating vector functions

Differentiability

A vector-valued function is differentiable at $t = a$ if the derivative exists, which is defined as:

$$\mathbf{r}'(a) = \lim_{h\to 0} \frac{\mathbf{r}(a+h) - \mathbf{r}(a)}{h}$$

The derivative of a vector-valued function gives the tangent vector to the curve at that point.

Definition 2.19 The **derivative $\mathbf{r}'$** of a vector function $\mathbf{r}$ is defined in the same manner as for real-valued functions:

$$\frac{d\mathbf{r}}{dt} = \mathbf{r}'(t) = \lim_{h\to 0} \frac{\mathbf{r}(t+h) - \mathbf{r}(t)}{h} = \lim_{\Delta t\to 0} \frac{1}{\Delta t}(\mathbf{r}(t+\Delta t) - \mathbf{r}(t))$$

◆

Theorem 2.20 *If $\mathbf{r}(t) = x(t)\mathbf{i} + y(t)\mathbf{j} + z(t)\mathbf{k} = \langle x(t),\ y(t),\ z(t)\rangle$ where x, y, and z are differentiable functions, then:*

$$\mathbf{r}'(t) = x'(t)\mathbf{i} + y'(t)\mathbf{j} + z'(t)\mathbf{k} = \langle x'(t),\ y'(t),\ z'(t)\rangle$$

Proof. This rule is easily proven by applying the component-wise limits directly into the fundamental definition of the derivative of vector functions. ■

If the points P and Q have position vectors $\mathbf{r}(t)$ and $\mathbf{r}(t+h)$, then the displacement vector $\vec{PQ}$ represents the vector $\mathbf{r}(t+h) - \mathbf{r}(t)$. As $h \to 0$, the vector $\mathbf{r}(t+h) - \mathbf{r}(t)$ approaches a vector that lies along the target trajectory's path.

The vector $\mathbf{r}'(t)$ is called the **tangent vector** to the curve defined by $\mathbf{r}(t)$ at P, provided that $\mathbf{r}'(t)$ exists and $\mathbf{r}'(t) \neq \mathbf{0}$. The tangent line to C at P is defined to be the line through P parallel to the tangent vector $\mathbf{r}'(t)$. (In a 2D plane, the slope of the line through P parallel to $\mathbf{r}'(t)$ evaluates to $\frac{y'(t)}{x'(t)}$).

The Unit Tangent Vector.

We define the **unit tangent vector** as:

$$\mathbf{T}(t) = \frac{\mathbf{r}'(t)}{|\mathbf{r}'(t)|}$$

Corollary 2.21 *If $\mathbf{r}'(t)$ exists, then we say $\mathbf{r}(t)$ is differentiable and write:*

$$\mathbf{r}'(t) = D_t\mathbf{r}(t) = \frac{d}{dt}\mathbf{r}(t)$$

Similarly, the second derivative $\mathbf{r}''(t)$ is defined as:

$$\mathbf{r}''(t) = x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k} = \langle x''(t),\ y''(t),\ z''(t)\rangle$$

Proof. Let $\mathbf{r}(t) = \langle x(t), y(t), z(t)\rangle$. By applying the component derivative theorem, the first derivative is computed component-by-component:

$$\frac{d}{dt}\mathbf{r}(t) = \left\langle \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right\rangle = \langle x'(t), y'(t), z'(t)\rangle$$

Since each component function $x'(t)$, $y'(t)$, and $z'(t)$ is a standard real-valued scalar function, we can differentiate a second time using standard single-variable calculus rules:

$$\begin{aligned}\mathbf{r}''(t) &= \frac{d}{dt}\left[\mathbf{r}'(t)\right] \\ &= \left\langle \frac{d}{dt}[x'(t)], \frac{d}{dt}[y'(t)], \frac{d}{dt}[z'(t)] \right\rangle \\ &= \langle x''(t), y''(t), z''(t)\rangle\end{aligned}$$

Expressing this result in terms of the standard unit basis vectors yields $x''(t)\mathbf{i} + y''(t)\mathbf{j} + z''(t)\mathbf{k}$. ■

Example 2.22 Let $\mathbf{r}(t) = \ln t\ \mathbf{i} + e^{-3t}\mathbf{j} + t^2\mathbf{k}$. Find the derivative $\mathbf{r}'(t)$ where $t > 0$.

Solution. We differentiate each scalar component function with respect to t:

$$\begin{aligned}\frac{d}{dt}(\ln t) &= \frac{1}{t} \\ \frac{d}{dt}(e^{-3t}) &= -3e^{-3t} \\ \frac{d}{dt}(t^2) &= 2t\end{aligned}$$

Combining these component results gives the derivative vector function:

$$\mathbf{r}'(t) = \frac{1}{t}\mathbf{i} - 3e^{-3t}\mathbf{j} + 2t\mathbf{k}$$

▲

Example 2.23 Let $\mathbf{r}(t) = (1 + t^3)\mathbf{i} + te^{-t}\mathbf{j} + \sin 2t\mathbf{k}$.

1. Find $\mathbf{r}'(t)$ and $\mathbf{r}''(t)$.
2. Find the unit tangent vector at the point where $t = 0$.

Solution.

1. Differentiating component-by-component using the power rule, product rule, and chain rule yields the first derivative:

$$\mathbf{r}'(t) = 3t^2\mathbf{i} + (e^{-t} - te^{-t})\mathbf{j} + 2\cos 2t\mathbf{k}$$

   Differentiating a second time gives the second derivative:

$$\mathbf{r}''(t) = 6t\mathbf{i} + (-e^{-t} - e^{-t} + te^{-t})\mathbf{j} - 4\sin 2t\mathbf{k} = 6t\mathbf{i} + (te^{-t} - 2e^{-t})\mathbf{j} - 4\sin 2t\mathbf{k}$$

2. To find the unit tangent vector $\mathbf{T}(0)$, we first evaluate the velocity derivative vector at $t = 0$:

$$\mathbf{r}'(0) = 3(0)^2\mathbf{i} + (e^0 - 0)\mathbf{j} + 2\cos(0)\mathbf{k} = \mathbf{j} + 2\mathbf{k} = \langle 0, 1, 2\rangle$$

Next, compute the magnitude of this tangent vector:

$$|\mathbf{r}'(0)| = \sqrt{0^2 + 1^2 + 2^2} = \sqrt{5}$$

Divide the vector by its magnitude to obtain the final unit tangent vector:

$$\mathbf{T}(0) = \frac{\mathbf{r}'(0)}{|\mathbf{r}'(0)|} = \frac{1}{\sqrt{5}}\mathbf{j} + \frac{2}{\sqrt{5}}\mathbf{k} = \left\langle 0, \frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}} \right\rangle$$

▲

Checkpoint 2.24 For the curve $\mathbf{r}(t) = \sqrt{t}\mathbf{i} + (2-t)\mathbf{j}$, find $\mathbf{r}'(t)$ and sketch the position vector $\mathbf{r}(1)$ and the tangent vector $\mathbf{r}'(1)$.

Solution. Differentiating each component with respect to t yields:

$$\mathbf{r}'(t) = \frac{1}{2\sqrt{t}}\mathbf{i} - \mathbf{j}$$

Evaluating both the original position vector and the derivative tangent vector at $t = 1$ gives:

$$\mathbf{r}(1) = \mathbf{i} + \mathbf{j} = \langle 1, 1 \rangle$$

$$\mathbf{r}'(1) = \frac{1}{2}\mathbf{i} - \mathbf{j} = \left\langle \frac{1}{2}, -1 \right\rangle$$

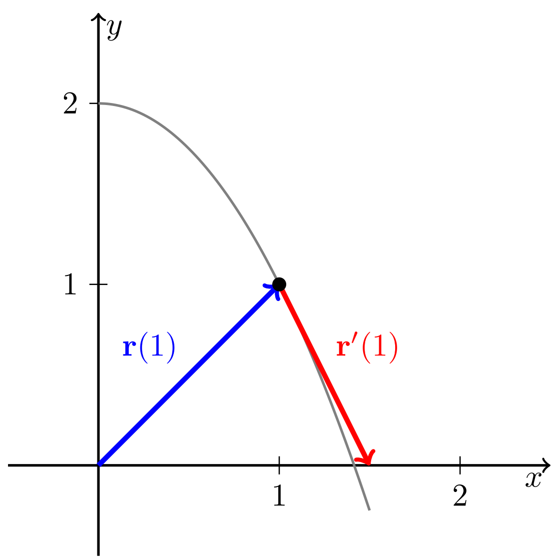


Figure 2.25 The position vector $\mathbf{r}(1)$ and the tangent vector $\mathbf{r}'(1)$ positioned on the curve.

Checkpoint 2.26 Find the parametric equations for the tangent line to the helix with the parametric equations:

$$x = 2\cos t, \quad y = \sin t, \quad z = t \quad \text{at the point } (0, 1, \pi/2)$$

Solution. First, find the parameter value t_0 that corresponds to the given point $(0, 1, \pi/2)$. Matching the components yields $z = t = \pi/2$. Testing this value reveals $2\cos(\pi/2) = 0$ and $\sin(\pi/2) = 1$, which confirms that $t_0 = \pi/2$.

Next, differentiate the vector function $\mathbf{r}(t) = \langle 2\cos t, \sin t, t \rangle$ to find the directional tangent vector:

$$\mathbf{r}'(t) = \langle -2\sin t, \cos t, 1 \rangle$$

Evaluating the direction vector at our parameter point $t_0 = \pi/2$ gives:

$$\mathbf{r}'(\pi/2) = \langle -2(1), 0, 1\rangle = \langle -2, 0, 1\rangle$$

Using the base point coordinates $(0, 1, \pi/2)$ and the direction components $\langle -2, 0, 1\rangle$, the parametric equations for the tangent line are:

$$x = -2t, \quad y = 1, \quad z = \frac{\pi}{2} + t$$

Checkpoint 2.27 Find the parametric equations for the tangent line to the twisted curve with the parametric equations:

$$x = t, \quad y = t^2, \quad z = t^3 \qquad \text{at the point } t = 2$$

Solution. Evaluate the base point on the curve at $t = 2$:

$$x_0 = 2, \quad y_0 = 2^2 = 4, \quad z_0 = 2^3 = 8 \implies (2, 4, 8)$$

Differentiate the component expressions to find the general velocity direction profile:

$$\frac{dx}{dt} = 1, \quad \frac{dy}{dt} = 2t, \quad \frac{dz}{dt} = 3t^2$$

Evaluating these derivatives at $t = 2$ gives the parallel directional components for our tangent line:

$$\left.\frac{dx}{dt}\right|_{t=2} = 1, \quad \left.\frac{dy}{dt}\right|_{t=2} = 4, \quad \left.\frac{dz}{dt}\right|_{t=2} = 12$$

Using the point $(2, 4, 8)$ and the direction components vector $\langle 1, 4, 12\rangle$, the parametric equations for the line are:

$$x = 2 + t, \quad y = 4 + 4t, \quad z = 8 + 12t$$

Definition 2.28 A curve given by a vector function $\mathbf{r}(t)$ on an interval I is called **smooth** if $\mathbf{r}'(t)$ is continuous and $\mathbf{r}'(t) \neq \mathbf{0}$, except possibly at the endpoints of I. ◆

Example 2.29 Determine whether the semicubical parabola $\mathbf{r}(t) = \langle 1 + t^3, t^2\rangle$ is smooth.

Solution. First, we compute the derivative vector function component-by-component:

$$\mathbf{r}'(t) = \langle 3t^2, 2t\rangle$$

Both component functions, $3t^2$ and $2t$, are polynomials, so $\mathbf{r}'(t)$ is continuous everywhere. Next, we determine if there are any parameter values where the derivative vector equals the zero vector:

$$\mathbf{r}'(t) = \langle 0, 0\rangle \implies 3t^2 = 0 \text{ and } 2t = 0$$

This system yields the single solution $t = 0$. Since $\mathbf{r}'(0) = \mathbf{0}$ at an interior point of the domain, the curve is **not smooth** at the origin (it features a sharp cusp). ▲

A curve like the semicubical parabola that is made up of a finite number of smooth pieces is called **piecewise smooth**.

Example 2.30 Determine whether the helix $\mathbf{r}(t) = \langle 2\cos t, \sin t, t\rangle$ is smooth.

Solution. We find the derivative vector function:

$$\mathbf{r}'(t) = \langle -2\sin t, \cos t, 1\rangle$$

The components are standard trigonometric and constant functions, meaning $\mathbf{r}'(t)$ is continuous for all real numbers.

For the vector to be the zero vector, all three components must simultaneously equal zero. However, the **k**-component is always $1 \neq 0$. Therefore, $\mathbf{r}'(t) \neq \mathbf{0}$ for all values of t. We conclude that the helix is a **smooth curve**. ▲

Theorem 2.31 *Suppose* $\mathbf{u}$ *and* $\mathbf{v}$ *are differentiable vector functions,* c *is a scalar, and* f *is a real-valued function. Then:*

1. $$\frac{d}{dt}[\mathbf{u}(t) + \mathbf{v}(t)] = \mathbf{u}'(t) + \mathbf{v}'(t)$$

2. $$\frac{d}{dt}[c\mathbf{u}(t)] = c\mathbf{u}'(t)$$

3. $$\frac{d}{dt}[f(t)\mathbf{u}(t)] = f'(t)\mathbf{u}(t) + f(t)\mathbf{u}'(t)$$

4. $$\frac{d}{dt}[\mathbf{u}(t)\cdot\mathbf{v}(t)] = \mathbf{u}'(t)\cdot\mathbf{v}(t) + \mathbf{u}(t)\cdot\mathbf{v}'(t)$$

5. $$\frac{d}{dt}[\mathbf{u}(t)\times\mathbf{v}(t)] = \mathbf{u}'(t)\times\mathbf{v}(t) + \mathbf{u}(t)\times\mathbf{v}'(t)$$

6. $$\frac{d}{dt}[\mathbf{u}(f(t))] = f'(t)\mathbf{u}'(f(t)) \quad \textit{(Chain Rule)}$$

Theorem 2.32 *If* $\mathbf{r}(t)$ *is differentiable and* $|\mathbf{r}(t)| = c$ *(a constant), then* $\mathbf{r}'(t)$ *is orthogonal to* $\mathbf{r}(t)$ *for all* t *in the domain of* $\mathbf{r}'(t)$*:*

$$\mathbf{r}(t)\cdot\frac{d\mathbf{r}}{dt} = 0$$

Proof. Since $|\mathbf{r}(t)| = c$, squaring both sides gives $|\mathbf{r}(t)|^2 = c^2$. Using the relationship between the magnitude and the dot product, we can rewrite this as:

$$\mathbf{r}(t)\cdot\mathbf{r}(t) = c^2$$

Differentiating both sides with respect to t using the dot product rule yields:

$$\mathbf{r}'(t)\cdot\mathbf{r}(t) + \mathbf{r}(t)\cdot\mathbf{r}'(t) = 0$$

Since the dot product is commutative, we combine the terms:

$$2\left(\mathbf{r}(t)\cdot\mathbf{r}'(t)\right) = 0 \implies \mathbf{r}(t)\cdot\frac{d\mathbf{r}}{dt} = 0$$

Because their dot product is zero, the position vector $\mathbf{r}(t)$ and the derivative vector $\mathbf{r}'(t)$ are orthogonal. ■

Chapter 3

MyOpenMath Interactive Activities

These files contain copies of the interactive MyOpenMath exercises that guided our learning for each topic. Special thanks to Professor Jason Hardin for sharing his MyOpenMath question bank.

Chapter 4

Chapter Exercises

This chapter contains exercises for students to practice the concepts learned in the course. Each exercise is designed to reinforce understanding and application of each chapter's topics.

Worksheet: Vectors and the Geometry of Space

1. When do directional line segments in the plane represent the same vector?

Solution. Directional line segments represent the same vector if and only if they are **equivalent**. Geometrically, this means they must have the exact same **magnitude** (length) and the exact same **direction**, regardless of where their initial points are located in the coordinate plane.

2. How do you find a vector's direction?

Solution. The direction of a non-zero vector $\vec{v}$ is found by computing its corresponding **unit vector**, denoted as $\vec{u}$. You find this by dividing the vector by its scalar magnitude:

$$\vec{u} = \frac{\vec{v}}{\|\vec{v}\|}$$

Alternatively, in two dimensions, the direction can be specified by the angle θ it makes with the positive horizontal x-axis, calculated using $\tan(\theta) = \frac{v_2}{v_1}$.

3. What geometric interpretation does the dot product have? Give an example.

Solution. Geometrically, the dot product of two vectors measures the product of their lengths multiplied by the cosine of the angle between them: $\vec{a} \cdot \vec{b} = \|\vec{a}\|\|\vec{b}\| \cos(\theta)$. It can be interpreted as the length of $\vec{a}$ multiplied by the signed scalar projection of $\vec{b}$ onto $\vec{a}$ ($\text{comp}_{\vec{a}}\, \vec{b}$).

Example: If two non-zero vectors are perpendicular to each other ($\theta = \pi/2$), their dot product is exactly 0 because $\cos(\pi/2) = 0$. This provides a tool to test for orthogonality.

4. Find the component form of the vector that is obtained by rotating $\langle 0, 1\rangle$ through an angle of $2\pi/3$ radians.

Solution. The initial vector $\vec{v} = \langle 0, 1\rangle$ points straight up along the positive y-axis, meaning its initial direction angle is $\theta_0 = \pi/2$. Rotating it by $2\pi/3$ updates its direction angle to:

$$\theta = \frac{\pi}{2} + \frac{2\pi}{3} = \frac{3\pi}{6} + \frac{4\pi}{6} = \frac{7\pi}{6}$$

Since the magnitude of $\langle 0, 1\rangle$ is 1, the component form of the rotated unit vector is:

$$\vec{v}_{\text{rot}} = \left\langle \cos\left(\frac{7\pi}{6}\right), \sin\left(\frac{7\pi}{6}\right)\right\rangle = \left\langle -\frac{\sqrt{3}}{2}, -\frac{1}{2}\right\rangle$$

5. Find the component form of the vector that is 2 units long in the direction of $4\vec{i} - \vec{j}$.

Solution. Let $\vec{w} = \langle 4, -1\rangle$. First, find the unit vector in the direction of $\vec{w}$ by calculating its magnitude:

$$\|\vec{w}\| = \sqrt{4^2 + (-1)^2} = \sqrt{16 + 1} = \sqrt{17}$$

The direction unit vector is $\vec{u} = \frac{1}{\sqrt{17}}\langle 4, -1\rangle$.

Scaling this direction vector to a length of 2 yields:

$$\vec{v} = 2\vec{u} = \frac{2}{\sqrt{17}}\langle 4, -1\rangle = \left\langle \frac{8}{\sqrt{17}}, -\frac{2}{\sqrt{17}}\right\rangle$$

6. Express the vector in terms of its length and direction:

$$\vec{v} = (-2\sin(t))\vec{i} + (2\cos(t))\vec{j} \quad \text{when} \quad t = \pi/2$$

Solution. First evaluate the vector at the parameter value $t = \pi/2$:

$$\vec{v} = \left(-2\sin\left(\frac{\pi}{2}\right)\right)\vec{i} + \left(2\cos\left(\frac{\pi}{2}\right)\right)\vec{j} = -2(1)\vec{i} + 2(0)\vec{j} = -2\vec{i} = \langle -2, 0\rangle$$

Now decompose the vector into its length and direction components:

- Length (magnitude): $\|\vec{v}\| = \sqrt{(-2)^2 + 0^2} = 2$
- Direction vector: $\vec{u} = \frac{\vec{v}}{\|\vec{v}\|} = \frac{\langle -2, 0\rangle}{2} = \langle -1, 0\rangle = -\vec{i}$

In factored form, the vector is modeled as $\vec{v} = 2(-\vec{i})$.

7. Find $\text{proj}_{\vec{v}}\,\vec{u}$ if $\vec{v} = 2\vec{i} + \vec{j} - \vec{k}$ and $\vec{u} = \vec{i} + \vec{j} - 5\vec{k}$.

Solution. Write both vectors in component form: $\vec{v} = \langle 2, 1, -1\rangle$ and $\vec{u} = \langle 1, 1, -5\rangle$. Next, calculate the dot product $\vec{u} \cdot \vec{v}$ and the squared norm $\|\vec{v}\|^2$:

$$\vec{u} \cdot \vec{v} = (1)(2) + (1)(1) + (-5)(-1) = 2 + 1 + 5 = 8$$

$$\|\vec{v}\|^2 = 2^2 + 1^2 + (-1)^2 = 4 + 1 + 1 = 6$$

Substitute these calculations into the vector projection formula:

$$\text{proj}_{\vec{v}}\,\vec{u} = \left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{v}\|^2}\right)\vec{v} = \frac{8}{6}\langle 2, 1, -1\rangle = \frac{4}{3}\langle 2, 1, -1\rangle = \left\langle \frac{8}{3}, \frac{4}{3}, -\frac{4}{3}\right\rangle$$

8. If $\|\vec{v}\| = 2$, $\|\vec{w}\| = 3$, and the angle between $\vec{v}$ and $\vec{w}$ is $\pi/3$, find $\|\vec{v} - 2\vec{w}\|$.

Solution. We can square the target norm to evaluate it using the algebraic properties of the dot product:

$$\begin{aligned}
\|\vec{v} - 2\vec{w}\|^2 &= (\vec{v} - 2\vec{w}) \cdot (\vec{v} - 2\vec{w}) \\
&= \vec{v} \cdot \vec{v} - 4(\vec{v} \cdot \vec{w}) + 4(\vec{w} \cdot \vec{w}) \\
&= \|\vec{v}\|^2 - 4(\|\vec{v}\|\|\vec{w}\| \cos(\theta)) + 4\|\vec{w}\|^2
\end{aligned}$$

Substitute the given magnitudes and the angle value into our expression:

$$\begin{aligned}
\|\vec{v} - 2\vec{w}\|^2 &= (2)^2 - 4\left(2 \cdot 3 \cdot \cos\left(\frac{\pi}{3}\right)\right) + 4(3)^2 \\
&= 4 - 4\left(6 \cdot \frac{1}{2}\right) + 4(9) \\
&= 4 - 12 + 36 = 28
\end{aligned}$$

Taking the square root gives the final length: $\|\vec{v} - 2\vec{w}\| = \sqrt{28} = 2\sqrt{7}$.

9. Find the volume of the parallelepiped determined by the vectors:

$$\vec{u} = \vec{i} + \vec{j} - \vec{k}, \quad \vec{v} = 2\vec{i} + \vec{j} + \vec{k}, \quad \vec{w} = -\vec{i} - 2\vec{j} + 3\vec{k}$$

Solution. The volume of a parallelepiped matches the absolute value of the scalar triple product, computed using a single 3×3 matrix determinant:

$$V = \left|\begin{vmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ -1 & -2 & 3 \end{vmatrix}\right|$$

Evaluate the matrix determinant by expanding along its top row:

$$\begin{aligned}
\begin{vmatrix} 1 & 1 & -1 \\ 2 & 1 & 1 \\ -1 & -2 & 3 \end{vmatrix} &= 1\begin{vmatrix} 1 & 1 \\ -2 & 3 \end{vmatrix} - 1\begin{vmatrix} 2 & 1 \\ -1 & 3 \end{vmatrix} + (-1)\begin{vmatrix} 2 & 1 \\ -1 & -2 \end{vmatrix} \\
&= 1(3 - (-2)) - 1(6 - (-1)) - 1(-4 - (-1)) \\
&= 1(5) - 1(7) - 1(-3) = 5 - 7 + 3 = 1
\end{aligned}$$

Taking the absolute value, the volume of the parallelepiped is $V = 1$ cubic unit.

10. Find the distance from the point to the line:

$$(2,2,0); \quad x=-t, \quad y=t, \quad z=-1+t$$

Solution. Let the target point be $S(2,2,0)$. From the parametric equations, we read a point on the line by setting $t=0$, giving $P(0,0,-1)$, and extract the line's direction vector: $\vec{v}=\langle -1,1,1\rangle$.

Construct the displacement vector from P to S:

$$\vec{PS}=\langle 2-0,2-0,0-(-1)\rangle=\langle 2,2,1\rangle$$

The formula for the distance from a point to a line is $d=\frac{\|\vec{PS}\times\vec{v}\|}{\|\vec{v}\|}$. First, compute the cross product:

$$\vec{PS}\times\vec{v}=\begin{vmatrix}\vec{i} & \vec{j} & \vec{k}\\ 2 & 2 & 1\\ -1 & 1 & 1\end{vmatrix}=(2-1)\vec{i}-(2-(-1))\vec{j}+(2-(-2))\vec{k}=\langle 1,-3,4\rangle$$

Now calculate the magnitudes:

$$\|\vec{PS}\times\vec{v}\|=\sqrt{1^2+(-3)^2+4^2}=\sqrt{1+9+16}=\sqrt{26}$$

$$\|\vec{v}\|=\sqrt{(-1)^2+1^2+1^2}=\sqrt{3}$$

Thus, the distance from the point to the line is $d=\frac{\sqrt{26}}{\sqrt{3}}=\sqrt{\frac{26}{3}}$.

11. Find the equation for the plane that passes through the point $(3,-2,1)$ and is normal to the vector $\vec{n}=2\vec{i}+\vec{j}+\vec{k}$.

Solution. Substitute the point coordinates $(3,-2,1)$ and the normal vector components $\langle 2,1,1\rangle$ directly into the standard scalar plane equation:

$$2(x-3)+1(y-(-2))+1(z-1)=0$$

Distribute and collect constant terms:

$$2x-6+y+2+z-1=0 \implies 2x+y+z-5=0$$

In standard linear form, the plane equation is $2x+y+z=5$.

12. Find the equation to the plane through the points:

$$P(1,-1,2), \quad Q(2,1,3), \quad R(-1,2,-1)$$

Solution. First, construct two directional vectors sharing vertex P that lie entirely within the plane:

$$\vec{PQ} = \langle 2-1, 1-(-1), 3-2\rangle = \langle 1,2,1\rangle$$

$$\vec{PR} = \langle -1-1, 2-(-1), -1-2\rangle = \langle -2,3,-3\rangle$$

The normal vector $\vec{n}$ is perpendicular to both vector components, found via their cross product:

$$\vec{n} = \vec{PQ} \times \vec{PR} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ -2 & 3 & -3 \end{vmatrix} = (-6-3)\vec{i} - (-3-(-2))\vec{j} + (3-(-4))\vec{k} = \langle -9,1,7\rangle$$

Using point $P(1,-1,2)$ and our calculated normal vector, set up the plane equation:

$$-9(x-1) + 1(y-(-1)) + 7(z-2) = 0 \implies -9x + 9 + y + 1 + 7z - 14 = 0$$

Simplifying the expression yields the linear equation: $-9x + y + 7z - 4 = 0$, or equivalently, $9x - y - 7z = -4$.

13. Find the distance from the point $(1,4,0)$ to the plane passing through $A(0,0,0)$, $B(2,0,-1)$, and $C(2,-1,0)$.

Solution. Construct two spanning vectors from the given points inside the plane:

$$\vec{AB} = \langle 2,0,-1\rangle \quad \text{and} \quad \vec{AC} = \langle 2,-1,0\rangle$$

Compute their cross product to find the normal vector direction $\vec{n}$:

$$\vec{n} = \vec{AB} \times \vec{AC} = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 2 & 0 & -1 \\ 2 & -1 & 0 \end{vmatrix} = (0-1)\vec{i} - (0-(-2))\vec{j} + (-2-0)\vec{k} = \langle -1,-2,-2\rangle$$

Since the plane passes through the origin $A(0,0,0)$, its linear equation simplifies to:

$$-1(x) - 2(y) - 2(z) = 0 \implies x + 2y + 2z = 0$$

Apply the point-plane distance formula using our target point $(1,4,0)$:

$$D = \frac{|1(1) + 2(4) + 2(0) + 0|}{\sqrt{1^2 + 2^2 + 2^2}} = \frac{|1+8|}{\sqrt{1+4+4}} = \frac{9}{\sqrt{9}} = \frac{9}{3} = 3$$

The exact distance from the point to the plane is 3 units.

14. Find the parametric equations for the line in which the planes $x + 2y + z = 1$ and $x - y + 2z = -8$ intersect.

Solution. The direction vector $\vec{v}$ of the intersection line is perpendicular to the normal vectors of both planes: $\vec{n}_1 = \langle 1, 2, 1\rangle$ and $\vec{n}_2 = \langle 1, -1, 2\rangle$. Compute their cross product:

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 1 & 2 & 1 \\ 1 & -1 & 2 \end{vmatrix} = (4 - (-1))\vec{i} - (2 - 1)\vec{j} + (-1 - 2)\vec{k} = \langle 5, -1, -3\rangle$$

To locate an initial point on the line, choose a convenient coordinate value, such as setting $z = 0$. This reduces the equations of the planes to a system of two variables:

$$x + 2y = 1 \quad \text{and} \quad x - y = -8$$

Subtracting the second equation from the first yields $3y = 9 \implies y = 3$. Substituting $y = 3$ back gives $x = -5$. This provides the point coordinates $(-5, 3, 0)$.

Combining our initial position point and the direction numbers yields the **parametric equations**:

$$x = -5 + 5t, \quad y = 3 - t, \quad z = -3t$$

15. Find a vector of magnitude 2 parallel to the line of intersection of the planes $x+2y+z-1 = 0$ and $x-y+2z+7 = 0$.

Solution. The direction vector $\vec{v}$ of the line of intersection is perpendicular to both normal vectors, $\vec{n}_1 = \langle 1, 2, 1\rangle$ and $\vec{n}_2 = \langle 1, -1, 2\rangle$. As calculated in Exercise 13, this cross product is:

$$\vec{v} = \vec{n}_1 \times \vec{n}_2 = \langle 5, -1, -3\rangle$$

To scale this direction to a specific length, first determine its unit vector by computing its magnitude:

$$\|\vec{v}\| = \sqrt{5^2 + (-1)^2 + (-3)^2} = \sqrt{25 + 1 + 9} = \sqrt{35}$$

The corresponding unit vector is $\vec{u} = \frac{1}{\sqrt{35}}\langle 5, -1, -3\rangle$.

Multiplying the direction unit vector by our target magnitude of 2 yields the final vector:

$$\vec{v}_{\text{final}} = \pm 2\vec{u} = \pm\frac{2}{\sqrt{35}}\langle 5, -1, -3\rangle = \pm\left\langle \frac{10}{\sqrt{35}}, -\frac{2}{\sqrt{35}}, -\frac{6}{\sqrt{35}} \right\rangle$$

Either the positive or negative scalar multiple satisfies the geometric condition.

Chapter 5

Homework

Chapter 6

Course Documents

MA-105-BL: Survey of Math

Worcester State University, DGCE, Summer 2026

Course Information

Instructor	Dr. Hansun To, Professor of Mathematics, hto1@worcester.edu.
Meeting Information	This is a **Hybrid Course**, combining online and in-person learning to provide a flexible and tailored experience. You'll have the opportunity to learn at your own pace using MyOpenMath, participate in Zoom discussions as needed, and attend in-person sessions for exams. *In-person attendance is required* for the following sessions: • Two midterm exams (1 hour and 30 minutes): ◦ (In person) Thursday, June 4th, from 03:30 PM to 05:00 PM ◦ (Online) Thursday, June 25th. (The exam will be available on Thursday, June 25th, at 08:00 AM. Please submit your work by 06:30 PM, including any necessary show work attachments.) • Final exam (three hours): ◦ (In person) Thursday, July 2nd, from 03:30 PM to 05:30 PM All in-person exams will be held in Sullivan room S-136. Please plan accordingly to ensure your attendance at these mandatory sessions.
Prerequisite	Math placement exam code 3 or higher, or a weighted high school GPA of 2.7 or higher within the past 3 years

Responsibility for Learning. Every student is responsible for their own learning. The instructor will make every effort to support you; however, if you need any help it is your responsibility to reach out with questions

Course Content. In this course we will learn to improve the level of quantitative awareness of students using familiar situations that provide a sense of purpose for studying mathematics. The objective is not to make mathematicians of the students, but to help them deal as comfortably as possible with an environment that increasingly makes use of quantitative reasoning. Topics will include voting and apportionment methods, personal finance, set theory, and probability theory.

Required Materials. A calculator is required for this course. There is no required textbook; notes will be provided in class (or in pdf format) and other materials will be freely available on the MyOpenMath platform.

Calculators. A scientific, non-graphing calculator capable of computing logarithmic and trigonometric values is required, though its use will be limited to only a handful of sections. I recommend the ***TI-30X IIS***, which you can usually find for around $15. Graphing calculators, phones, tablets and any unauthorized devices are never allowed!

MyOpenMath. MyOpenMath (MOM) is a free, open source, online course management system designed for mathematics which we will use extensively in this course. It is available at https://www.myopenmath.com/ . You will need to create an account and register for our course on MyOpenMath to get started; we will do this as a group during the first class meeting. All course documents and materials will be posted on MyOpenMath; and all homework will be completed through MyOpenMath.

Assignments and Grading

Homework. Practice is a primary component of the mathematical learning process. For this reason, homework sets will be assigned on MyOpenMath after most class meetings. Assignments will always be due on Wednesdays by 11:59pm. This means you will often have two homework sets due on Wednesday, one which opened on the previous Monday and the other which opened on the previous Wednesday. While the number of problems in each homework set will vary, your score will be considered as a percentage and each set will be weighted equally when computing your homework grade. The problems on the homework sets are sometimes meant to deepen or extend the understanding you have gained from the class lecture and discussion; you should expect to find some of the exercises difficult.

The online homework is set up so you can try most problems two times, then the correct answer will be shown (there may be some questions in which you have only one try, this will be indicated in the homework set). If you miss a problem and want to improve your score, you can click the ``Try a similar problem'' link, which will give you a new question of the same type. You can keep on working on versions of a question until you get it correct. ***You must earn at least 50% on an assignment for your score to count towards your grade; any set in which you earn less than 50% will be counted as a 0.*** Assignments will remain available in "Practice" mode after their due date to provide additional practice of concepts; new versions of problems can be generated and answered, but scores will not be saved. ***Be aware, opening an online homework assignment in "Practice" mode will disqualify***

you from being able to redeem a LatePass on that assignment! If you are having trouble on a problem, you may use the "Message Instructor" link, which will allow you to send me a message through MyOpenMath; the specific problem you are working on will be automatically included in the message. I can then reply to the message with some guidance. These messages are internal to MyOpenMath and not email, so my reply will go to your messages when you login to MyOpenMath. I will check my messages on MyOpenMath at least once per day. Some problems may have a link to a video solution of a similar problem.

To account for any unexpected emergencies or technical issues that may occur, we will utilize the LatePass feature in MyOpenMath. A LatePass can be redeemed for an automatic 48-hour extension on any online homework set which has NOT been viewed in "Practice" mode. To use a LatePass, click the "Use LatePass" button next to the appropriate assignment on MyOpenMath and the platform will automatically extend the due date for that assignment to 48 hours after the original due date. Each student will be given 8 LatePasses to use as they wish throughout the semester. Note a maximum of two LatePasses can be used on any single assignment. If circumstances require a more substantial extension, contact your instructor as soon as possible to discuss this possibility.

Exams. There will be four exams in this course, one of which will be a cumulative final exam. Tentative dates are June 4, June 25, May 6, and July 2.

Grade Allocation. Your course grade will be determined by the following distribution. ***You must get at least a 50% on the final exam in order to pass the class***.

Assignment	**Percentage**
MyOpenMath Online Homework	10%
MyOpenMath Online Interactive Activities	10%
MyOpenMath Online Quizzes	10%
Exam 1	20%
Exam 2 (online)	20%
Final Exam	30%

Grading Scale. Letter grades will be assigned as follows. Grades will not be curved.

Letter Grade	**Percent Needed**
A	93%
A-	90%
B+	87%
B	83%
B-	80%
C+	77%
C	73%
C-	70%
D+	67%
D	63%
D-	60%
E	Below 60%

Course Policies

Email. I try to respond to emails and MyOpenMath messages within 24 hours. You are encouraged to email/message regarding the homework; however, emails sent within 24 hours of a due date may not get a response before the deadline. So do not wait until the last minute to attempt the assignments!

Studying. To succeed in this course, you will need to do more than simply attend class. Mathematics is a subject which is learned by doing, meaning it is important to work problems outside of class to fully understand the concepts we cover. The assigned homework is the minimum amount you should be completing; if you feel like you don't fully understand a concept, ask me to guide you to some additional practice problems. A general rule for studying (in all college courses) is for each hour of class per week you spend a minimum of two hours outside of class on homework, studying, reading the text, etc.

Since this is a hybrid course and it is three credit hours, you should plan to spend at least six hours per week working on course material for a regular semester. Since this is a 7-week summer course, that time is doubled to 12 hours per week.

Academic Integrity and AI Usage. The WSU Mathematics Department does not tolerate academic dishonesty! For an explanation of what is considered "academic dishonesty" at WSU, please see the policy included in the 2025-2026 Undergraduate Catalog. All instructors in the WSU Mathematics Department are required to report all incidents of academic dishonesty observed, detected, or otherwise determined to have occurred to the WSU Academic Central File. WSU requires some penalty be imposed for every reported violation of the academic honesty policy. All incidents of academic dishonesty in this course will result in (at least) a score of 0 on that assignment; additional penalties will be imposed at the instructor's discretion and/or as required by the Mathematics Department Academic Honesty Policy.

Artificial Intelligence (AI) Policy: Artificial intelligence (AI) tools (ChatGPT and similar platforms) may be used for *study support and practice only.* Acceptable uses include reviewing concepts, generating practice problems, checking understanding of definitions, or exploring alternative explanations of course material. AI tools are not permitted on any graded assignments, quizzes, or examinations unless explicitly authorized by the instructor. This includes, but is not limited to, using AI to generate solutions, explanations, written work, computations, graphs, or code that is submitted for credit. All submitted work must reflect the student's own understanding and effort. Reliance on AI-generated responses in place of independent problem-solving undermines the learning objectives of this course and is not allowed. Students are responsible for ensuring that any work they submit complies with this policy. Use of AI in violation of this policy will be treated as a violation of the Academic Honesty Policy and may result in penalties including a score of zero on the assignment, quiz, or exam, and additional disciplinary action as required by university policy.

Instructor Rights. ***As the instructor of this course, I reserve the right to make changes to the above course policies at any time. All changes will be announced in class.***

Student Learning Outcomes

Students will be able to:

1. Students will be able to communicate their mathematical reasoning.
2. Financial Management: Students will be able to calculate simple and compound interest, loan installments and mortgage payments.
3. Probability Theory: Students will be able to determine sample spaces for experiments, calculate probabilities with fundamental counting principles, permutations and combinations.
4. Voting Methods: Students will be able to discuss the pros and cons of several voting methods and how they can be manipulated by a knowledgeable individual.

LASC

This is a Quantitative Reasoning (QR) course in the Liberal Arts and Sciences Curriculum (LASC).

The course addresses the following LASC Quantitative Reasoning objectives:

1. Acquaint students with formal systems, procedures, and sequences of operations.
2. Strengthen students' understanding of variables and functions.
3. Apply mathematical techniques to the analysis and solution of real-life problems.
4. Emphasize the importance of accuracy, including precise language and careful definitions of mathematical concepts.
5. Understand both the underlying principles and practical applications of one or more fields of mathematics.
6. Develop an understanding of and facility with statistical analysis an analysis, including understanding of its applications and limitations.

and the following LASC overarching objectives:

1. Communicate effectively orally and in writing.
2. Understand and employ quantitative and qualitative reasoning.
3. Apply skills in critical thinking.
4. Understand the roles of science and technology in our modern world.
5. Become socially responsible agents in the world.
6. Make connections across courses and disciplines.
7. Develop as healthy individuals--physically, emotionally, socially, ethically, and intellectually.

Course Schedule

Here is a tentative schedule of topics. We will try to stay on schedule as much as possible. If class is canceled for any reason, then the schedule will be adjusted to accommodate this cancelation. This could include a rearrangement of topics, removing a "review" day, or skipping a topic entirely, depending on the timing of the cancelation. Any changes to this schedule will be announced in class and via email.

Table 6.1 Course Schedule and Exam Dates

Unit	Topics	Due Dates
1	Voting methods, apportionment	Fri 5/22
2	Simple interest, compound interest	Wed 5/27
3	Annuities, installment loans ***(Quiz 1)***	Wed 5/27
4	Mortgages, credit cards	Wed 6/3
	In-Person Exam 1 Thu 6/4 (3:30-5:00pm) (S-136)	
5	Set basics, subsets	Wed 6/10
6	Set operations, surveys ***(Quiz 2)***	Wed 6/10
7	Intro to probability, events involving “not”, “or”, “and”	Wed 6/17
8	Counting, permutations, combinations ***(Quiz 3)***	Wed 6/17
	Online Exam 2 (Thu 6/25)	
9	Probability with counting, odds, conditional	Fri 6/26
	In-Person FINAL EXAM Thu 7/2 (3:30-6:30pm) (S-136)	